

$$\mathcal{L} = i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi$$

ψ : Dirac field

$$\bar{u}_\pm(p)(\not{p} + m) = 0$$

$$\not{p} = \begin{pmatrix} 0 & p_{\alpha\dot{\beta}} \\ p_{\dot{\alpha}\beta} & 0 \end{pmatrix}$$

$$p_{\alpha\dot{\beta}} = p_\mu (\sigma^\mu)_{\alpha\dot{\beta}} = \begin{pmatrix} -p^0 + p^3 & p^1 - i p^2 \\ p^1 + i p^2 & -p^0 - p^3 \end{pmatrix}$$

$$\sigma^\mu = (1, \sigma^i), \quad \bar{\sigma}^\mu = (1, -\sigma^i)$$

$$\det(p_{\alpha\dot{\beta}}) = -p^\mu p_\mu = m^2$$

• When $m=0$:

$$\not{p} u_\pm(p) = 0$$

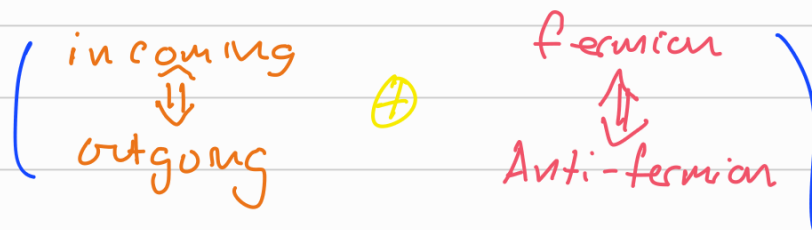
$$\bar{u}_\pm(p) \not{p} = 0$$

$u_\pm(p)$ and $\bar{u}_\pm(p)$: Outgoing
(Anti-ferm) (ferm)

v : Anti-ferm

u : fermion

Crossing-Symmetry: exchanges



$$\psi = \begin{pmatrix} \phi_\alpha \\ \chi_{\dot{\alpha}} \end{pmatrix}$$

$\bar{u}$: outgoing fermion
 u : incoming fermion

v : outgoing Anti-fermions
 $\bar{v}$: incoming Anti-fermions

• Note this symmetry flips helicity

$$u_\pm = v_\mp \quad \text{and} \quad \bar{u}_\pm = \bar{v}_\mp$$

$$u_+(p) = \begin{pmatrix} |p\rangle_\alpha \\ 0 \end{pmatrix}, \quad v_-(p) = \begin{pmatrix} 0 \\ |p\rangle^{\dot{\alpha}} \end{pmatrix}$$

$$\bar{u}_-(p) = (0, \langle p|_{\dot{\alpha}}), \quad \bar{u}_+(p) = (\langle p|^{\alpha}, 0)$$

$$p^{\dot{\alpha}\beta} |p\rangle_\alpha = 0$$

$$\langle p|^{\beta} p_{\beta\dot{\alpha}} = 0$$

$$p^{\alpha\dot{\alpha}} |P\rangle_{\dot{\alpha}} = 0$$

$$\langle P|_{\dot{\alpha}} p^{\dot{\alpha}\beta} = 0$$

$$\boxed{\begin{aligned} [P]^\alpha &= \zeta^{\alpha\beta} [P]_\beta \\ |P\rangle_{\dot{\alpha}} &= \zeta^{\dot{\alpha}\dot{\beta}} \langle P|_{\dot{\beta}} \end{aligned}}$$

$$\begin{array}{ccc} \psi^\alpha \psi_\alpha & \nearrow & SU(2) \\ \bar{\psi}_{\dot{\alpha}} \bar{\psi}^{\dot{\alpha}} & \searrow & \overline{SU(2)} \end{array}$$

Reality Conditions: The two massless Weyl eqs

$$\not{p} \psi_{\pm}(p) = 0$$

$$\bar{\psi}_{\pm}(p) \not{p} = 0$$

are related to each other via Dirac conj.

(i) Crossing Symmetry

$$\bar{\psi}_{\pm} = \bar{\psi}_{\mp}$$

$$\rightarrow \bar{\psi}_{\pm} \not{p} = 0 \Rightarrow \bar{\psi}_{\mp} \not{p} = 0$$

(ii) Apply Dirac conj.

$$\begin{aligned} (\bar{\psi}_{\mp} p_m \gamma^m)^c &= (\bar{\psi}_{\mp} \not{p})^\dagger \beta = \not{p}^\dagger (\psi^\dagger \beta)^\dagger \beta \\ &= \not{p}^\dagger \beta^\dagger \psi \beta = 0 \end{aligned}$$

$$\left\{ \begin{aligned} \{\gamma^m, \gamma^n\} &= -2\eta^{mn}, & (\gamma^0)^2 &= 1 \end{aligned} \right.$$

$$(\gamma^m)^\dagger = \gamma^0 \gamma^m \gamma^0$$

$$= \not{p} \beta \psi = 0$$

• Which holds only if $p_m \in \mathbb{R}^4$ (Real-valued)

$$\boxed{\begin{aligned} p_m \text{ real: } [P]^\alpha &= (|P\rangle_{\dot{\alpha}})^* \\ \langle P|_{\dot{\alpha}} &= ([P]_\alpha)^* \end{aligned}}$$

$$\begin{aligned} \text{ex: } \bar{\psi}_+ &= \begin{pmatrix} [P]_\alpha \end{pmatrix}^\dagger \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = (([P]_\alpha)^*, 0) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= (0, ([P]_\alpha)^*) \end{aligned}$$

$$L = \pi_- = (0, \langle p | \dot{\alpha} \rangle) \quad \langle p | \dot{\alpha} \rangle = (|p\rangle \dot{\alpha})^*$$

This is simply $(\psi_\alpha)^* = \bar{\psi}_\alpha$

* Spacetime Completeness relations ($m=0$)

$$\pi_- \pi_- + \pi_+ \pi_+ = -\not{p}$$

$$\begin{pmatrix} |p\rangle_\alpha \\ 0 \end{pmatrix} \otimes (0, \langle p | \dot{\alpha} \rangle) + \begin{pmatrix} 0 \\ |p\rangle^{\dot{\alpha}} \end{pmatrix} \otimes \begin{pmatrix} 0 \\ \langle p | \alpha \rangle \end{pmatrix}$$

$$\begin{pmatrix} 0 & |p\rangle_\alpha \langle p | \dot{\alpha} \\ |p\rangle^{\dot{\alpha}} \langle p | \alpha & 0 \end{pmatrix} = -\not{p} = - \begin{pmatrix} 0 & p_{\alpha\dot{\alpha}} \\ p^{\dot{\alpha}\alpha} & 0 \end{pmatrix}$$

$$\therefore -\not{p} = |p\rangle \langle p| + |p\rangle \langle p|$$

$$\begin{aligned} p_{\alpha\dot{\alpha}} &= -|p\rangle_\alpha \langle p | \dot{\alpha} \\ p^{\dot{\alpha}\alpha} &= -|p\rangle^{\dot{\alpha}} \langle p | \alpha \end{aligned}$$

* Let A be a 2×2 matrix, then if $\det(A) = 0$
 then $A_{\mu\nu} = -\lambda_\mu \tilde{\lambda}_\nu$
 • $\det(p_{\alpha\dot{\alpha}}) = m^2 = 0$

ex: Here we calculate an explicit form for $|p\rangle$ and $|p\rangle$

• Let $p^\mu = (E, E \sin(\theta) \cos(\phi), E \sin(\theta) \sin(\phi), E \cos(\theta))$
 i.e. p^μ be some arbitrary momenta in spherical coordinates

$$p_{\alpha\dot{\alpha}} = p_\mu (\sigma^\mu)_{\alpha\dot{\alpha}} = \begin{pmatrix} -E + E \cos\theta & E \sin\theta \cos\phi - i E \sin\theta \sin\phi \\ E \sin\theta \cos\phi + i E \sin\theta \sin\phi & -E - E \cos\theta \end{pmatrix}$$

$$p_{ii} = E(\cos\theta - 1)$$

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} E(\cos\theta - 1) & E \sin\theta e^{-i\phi} \\ E \sin\theta e^{i\phi} & -E(1 + \cos\theta) \end{pmatrix} = \begin{pmatrix} -2E \sin^2(\theta/2) & E \sin\theta e^{-i\phi} \\ E \sin\theta e^{i\phi} & -2E \cos^2(\theta/2) \end{pmatrix}$$

outgoing fermion $h = +1/2$

outgoing fermion $h = -1/2$

outgoing Antifermion $h = +1/2$

$$\bar{u}_+(p) \leftrightarrow ([p]^\alpha, 0)$$

$$\bar{u}_-(p) \leftrightarrow (0, \langle p | \dot{\alpha})$$

$$v_+(p) \leftrightarrow \begin{pmatrix} [p]_\alpha \\ 0 \end{pmatrix}$$

$$\text{outgoing Antifermion } h = -1/2 \quad v_-(p) \leftrightarrow \begin{pmatrix} 0 \\ |p\rangle_{\dot{\alpha}} \end{pmatrix}$$

- Positive helicity of an outgoing particle
 $[p]$ and $|p\rangle$
- Negative helicity of an outgoing particle
 $\langle p|$ and $|p\rangle$

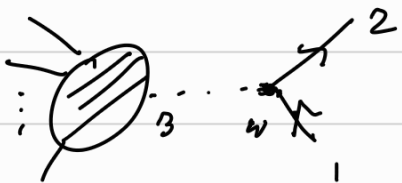
* Take all external particles to be outgoing



Yukawa Theory: $\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - \frac{1}{2}(\partial\phi)^2 + g\phi\bar{\psi}\psi$

Interaction term yields $i\int d^4x \mathcal{L}_I$

$$\langle \Omega | \phi_1 \bar{\psi}_2 \psi_3 \int d^4x \phi \bar{\psi} \psi | \Omega \rangle$$



$$= (ig) \bar{u}_{h_1}(p_1) v_{h_2}(p_2) \frac{-i}{(p_1 + p_2)^2} \phi(x_{\text{rest}})$$

$$(h_1 = +, h_2 = -)$$

$$\bar{u}_+ v_- = ([p_1]^\alpha, 0) \begin{pmatrix} 0 \\ |p_2\rangle_{\dot{\alpha}} \end{pmatrix} = 0$$

$(h = -, h = -)$

$$\bar{u}_- v_- = (0, \langle p_1 | \dot{\alpha}) \begin{pmatrix} 0 \\ |p_2\rangle_{\dot{\alpha}} \end{pmatrix} = \langle p_1 | \dot{\alpha} | p_2 \rangle_{\dot{\alpha}}$$

$$\equiv \langle p_1 p_2 \rangle$$

* For two light-like vectors p^μ, q^μ

$$\langle pq \rangle \equiv \langle p|_{\dot{\alpha}} |q\rangle^{\dot{\alpha}} \quad | [pq] = [p]^{\alpha} |q]_{\alpha}$$

Note:

$$\begin{aligned} \langle pq \rangle &= \langle p|_{\dot{\alpha}} |q\rangle^{\dot{\alpha}} = \epsilon_{\dot{\alpha}\dot{\beta}} |p\rangle^{\dot{\beta}} |q\rangle^{\dot{\alpha}} \\ &= -\epsilon_{\dot{\beta}\dot{\alpha}} |q\rangle^{\dot{\alpha}} |p\rangle^{\dot{\beta}} \\ &= -\langle q|_{\dot{\beta}} |p\rangle^{\dot{\beta}} \end{aligned}$$

and so

$$\boxed{\langle pq \rangle = -\langle qp \rangle} \quad | \quad \boxed{[pq] = -[qp]}$$

• For real momenta:

$$\boxed{[pq]^* = \langle qp \rangle}$$

$$\bullet \quad [p| : \mathcal{SU}(2)$$

$$\langle p| : \overline{\mathcal{SU}(2)}$$

$$\text{Claim: } \boxed{\langle pq \rangle [pq] = 2 p \cdot q = (p+q)^2}$$



$$\text{proof: } \text{Tr}(\sigma^\mu \bar{\sigma}^\nu) = -2\eta^{\mu\nu}$$

$$\begin{aligned} &\langle p|_{\dot{\alpha}} |q\rangle^{\dot{\alpha}} (-[q]^{\alpha} |p]_{\alpha}) \\ &= -q^{\dot{\alpha}\alpha} p_{\alpha\dot{\alpha}} \end{aligned}$$

$$= -q_\mu (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} p_\nu (\sigma^\nu)_{\alpha\dot{\alpha}}$$

$$= -q_\mu p_\nu \text{Tr}(\bar{\sigma}^\mu \sigma^\nu)$$

$$= -q_\mu p_\nu (-2\eta^{\mu\nu})$$

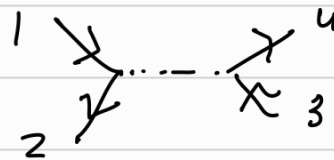
$$= 2 q \cdot p$$

$$\bullet \text{ Since } (p+q)^2 = \overset{\rightarrow 0}{p^2} + 2 p \cdot q + \overset{\rightarrow 0}{q^2} = 2 p \cdot q$$

$$\textcircled{11} \dots \frac{1}{\gamma} = ig \langle 12 \rangle \frac{-i}{\langle 12 \rangle [12]} \times (\text{rest}) = g \frac{1}{[12]} (\text{rest})$$

* 4-fermion tree amplitude

$$A_4(\bar{f}^{h_1} f^{h_2} \bar{f}^{h_3} f^{h_4})$$



1 and 3 are anti-fermions since they are incoming

$$\begin{array}{c} 1 \\ \nearrow \\ \text{---} \\ \nwarrow \\ 2 \end{array}
 \begin{array}{c} 4 \\ \nwarrow \\ \text{---} \\ \nearrow \\ 3 \end{array}
 = ig \bar{u}_4 v_3 \frac{-i}{(p_1 + p_2)^2} ig \bar{u}_2 v_1$$

• Particles 1 and 2 must have the same helicity. Same for 3 & 4.

$$h_1 = h_2 = -1/2$$

$$h_3 = h_4 = +1/2$$

• t and u-channels would vanish

$$= (ig)^2 (-i) \frac{[43] \langle 21 \rangle}{\langle 12 \rangle [12]} = ig^2 \frac{[34]}{[12]}$$

$$iA_4(\bar{f} f^+ \bar{f} f^+) = ig^2 \frac{[34]}{[12]}$$

Note: $\langle 12 \rangle [12] = (p_1 + p_2)^2 = (p_3 + p_4)^2 = \langle 34 \rangle [34]$
 $\langle 12 \rangle / \langle 34 \rangle = [34] / [12]$

$$A_4(\bar{f} f^+ \bar{f} f^+) = g^2 \frac{[34]}{[12]} = g^2 \frac{\langle 12 \rangle}{\langle 34 \rangle}$$

ex:

$$iA_4(f, \bar{f}^{h_2}, f^{h_3}, \bar{f})$$

$$\langle 0 | T \phi \bar{\psi} \psi \phi | 0 \rangle \sim (ig)^2 \int d^4 w d^4 z \phi_1 \phi_w \phi_2 \phi_z \bar{\psi} \psi \bar{\psi} \psi$$

$$= \begin{array}{c} 1 \text{---} \\ \nwarrow \\ 2 \end{array} \begin{array}{c} 4 \text{---} \\ \nwarrow \\ \text{---} \\ \nearrow \\ 3 \end{array} + \begin{array}{c} 4 \text{---} \\ \nwarrow \\ \text{---} \\ \nearrow \\ 3 \end{array} \begin{array}{c} 1 \text{---} \\ \nwarrow \\ 2 \end{array}$$

$$= ig \bar{u}_3 \frac{-i}{(p_1 + p_2)^2} v_2 (ig) + ig \bar{u}_2 \frac{-i}{(p_2 + p_4)^2} ig v_3$$

$$= -ig^2 \bar{u}_3 \frac{(p_1 + p_2)}{(p_1 + p_2)^2} v_2 - ig^2 \bar{u}_2 \frac{(p_2 + p_4)}{(p_2 + p_4)^2} v_3$$

Since $\not{p}\not{p} = -p^2$

Proof: $p_\mu \gamma^\mu p_\nu \gamma^\nu = -p_\mu \gamma^\nu \gamma^\mu p_\nu - 2p^2$

$2\not{p}\not{p} = -2p^2 \Rightarrow \not{p}\not{p} = -p^2$

Now, $\bar{u}_3 \gamma^\mu v_2 = (A, B) \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix}$

$= (A, B) \begin{pmatrix} \sigma^\mu D \\ \bar{\sigma}^\mu C \end{pmatrix} = A \sigma^\mu D + B \bar{\sigma}^\mu C$

So, we have a non zero product iff the spinors have opposite helicity

$\bar{u}_\pm(p_3) \gamma^\mu v_\pm(p_2) = 0$

$\bar{u}_-(p_3) \gamma^\mu v_+(p_2) = [3]_{\dot{\alpha}} \bar{\sigma}^{\mu \dot{\alpha} \alpha} [2]_{\alpha}$

$[3]_{\dot{\alpha}} \bar{\sigma}^{\mu \dot{\alpha} \alpha} [2]_{\alpha} \equiv [3] \gamma^\mu [2]$

$\bar{u}_+(p_3) \gamma^\mu v_-(p_2) = [3] \gamma^\mu [2]$

Same helicity: $[p] \gamma^\mu [k] = [p] \gamma^\mu [k] = 0$

* $[k] \gamma^\mu |p\rangle = \langle p| \gamma^\mu |k]$

$[k] \gamma^\mu |p\rangle^* = \langle p| \gamma^\mu |k\rangle$



Proof: $[k]^\alpha \sigma^\mu_{\alpha \dot{\alpha}} |p\rangle^{\dot{\alpha}} = \epsilon^{\alpha\beta} [k]_\beta \sigma^\mu_{\alpha \dot{\alpha}} \epsilon^{\dot{\alpha}\dot{\beta}} \langle p|_{\dot{\beta}}$

$= \langle p|_{\dot{\beta}} \epsilon^{\alpha\beta} \sigma^\mu_{\alpha \dot{\alpha}} \epsilon^{\dot{\alpha}\dot{\beta}} [k]_\beta$

$= \langle p|_{\dot{\beta}} (\bar{\sigma}^\mu)^{\dot{\beta} \beta} [k]_\beta$

$\therefore [k] \gamma^\mu |p\rangle = \langle p| \gamma^\mu |k]$

likewise $[k] \gamma^\mu |p\rangle^* = ([k]^\alpha \sigma^\mu_{\alpha \dot{\alpha}} |p\rangle^{\dot{\alpha}})^*$

$= |k\rangle^{\dot{\alpha}} \langle p|_{\alpha} \sigma^\mu_{\alpha \dot{\alpha}}$

$= \langle p|_{\alpha} \sigma^\mu_{\alpha \dot{\alpha}} |k\rangle^{\dot{\alpha}} = \langle p| \gamma^\mu |k\rangle$

* Now, we define: $\langle P | P | K \rangle \equiv P_\mu \langle P | \gamma^\mu | K \rangle$
 P_μ and K_μ are lightlike but P is not assumed to be
 no If P is like like ($P^2 = 0$) then

$$\langle P | P | K \rangle = \langle P | \dot{\alpha} P^{\dot{\alpha} \alpha} | K \rangle_\alpha = \langle P | \dot{\alpha} (-|P\rangle^{\dot{\alpha}} [P]^\alpha | K \rangle_\alpha) \\ = -\langle P P \rangle [P K]$$

* Fierz Identity:

 $\langle 1 | \gamma^\mu | 2 \rangle \langle 3 | \gamma_\mu | 4 \rangle = 2 \langle 13 \rangle [24]$ 

Proof:

$$\begin{aligned} \langle 1 | \gamma^\mu | 2 \rangle \langle 3 | \gamma_\mu | 4 \rangle &= \langle 1 | \dot{\alpha} \bar{\sigma}^{\mu \dot{\alpha} \alpha} | 2 \rangle_\alpha [4]^\beta \sigma_{\mu \beta \dot{\beta}} | 3 \rangle^{\dot{\beta}} \\ &= [2]^\alpha \sigma^{\mu}_{\alpha \dot{\alpha}} | 1 \rangle^{\dot{\alpha}} [4]^\beta \sigma_{\mu \beta \dot{\beta}} | 3 \rangle^{\dot{\beta}} \\ (\bar{\sigma}^\mu)_{\alpha \dot{\alpha}} (\sigma_\mu)_{\beta \dot{\beta}} &= -2 \epsilon_{\alpha \beta} \epsilon_{\dot{\alpha} \dot{\beta}} \\ (-2 \epsilon_{\alpha \beta} \epsilon_{\dot{\alpha} \dot{\beta}}) [2]^\alpha | 1 \rangle^{\dot{\alpha}} [4]^\beta | 3 \rangle^{\dot{\beta}} \\ &\quad - \epsilon_{\dot{\alpha} \dot{\beta}} | 1 \rangle^{\dot{\alpha}} = \langle 1 | \dot{\beta} \\ [2]^\alpha \epsilon_{\alpha \beta} &= \epsilon^{\alpha \gamma} \epsilon_{\alpha \beta} [2]_\gamma = -\delta^\gamma_\beta [2]_\gamma \\ &= -[2]_\beta \\ &= 2(-[2]_\beta) \langle 1 | \dot{\beta} | 3 \rangle^{\dot{\beta}} [4]^\beta \end{aligned}$$

$$\therefore \langle 1 | \gamma^\mu | 2 \rangle \langle 3 | \gamma_\mu | 4 \rangle = -2 \langle 13 \rangle [42] \\ = 2 \langle 13 \rangle [24]$$

Claim: $\langle K | \gamma^\mu | K \rangle = 2 K^\mu$ and $\langle K | P | K \rangle = 2 P \cdot K$

Proof: $\langle K | \dot{\alpha} (\bar{\sigma}^\mu)^{\dot{\alpha} \alpha} | K \rangle_\alpha = (\bar{\sigma}^\mu)^{\dot{\alpha} \alpha} | K \rangle_\alpha \langle K | \dot{\alpha} \\ = -(\bar{\sigma}^\mu)^{\dot{\alpha} \alpha} K_{\alpha \dot{\alpha}} = -(\bar{\sigma}^\mu)^{\dot{\alpha} \alpha} K_\gamma (\sigma^\gamma)_{\alpha \dot{\alpha}} \\ = -K_\gamma \text{Tr}(\bar{\sigma}^\mu \sigma^\gamma)$

Since

$$\{\sigma^\mu, \bar{\sigma}^\nu\} = -2\eta^{\mu\nu} \mathbb{1}$$

$$2 \text{Tr}(\sigma^{\mu\nu} \bar{\sigma}^{\mu\nu}) = -2\eta^{\mu\nu} (2)$$

$$\therefore \text{Tr}(\sigma^{\mu\nu} \bar{\sigma}^{\mu\nu}) = -2\eta^{\mu\nu}$$

Plugging this in

$$\langle n | \sigma^{\mu} | k \rangle = (-2\eta^{\mu\nu})(-k_{\nu}) = 2k^{\mu} \quad \checkmark$$

Multiplying this by P_{μ}

$$\langle k | P | k \rangle = 2k \cdot P \quad \checkmark$$

Notation: $\langle 1 | P_3 | 4 \rangle = \langle 1 | 3 | 4 \rangle = P_{3\mu} \langle 1 | \sigma^{\mu} | 4 \rangle$

$$\langle 2 | 1+5 | 4 \rangle = \langle 2 | (P_1 + P_5) | 4 \rangle$$

* Now, going back to $iA_4(\phi \bar{f}^{\mu_2} f^{\mu_3} \phi)$

$$= \begin{array}{c} 1 \\ \vdots \\ 2 \end{array} \begin{array}{c} \nearrow \\ \text{---} \\ \searrow \end{array} \begin{array}{c} 3 \\ \vdots \\ 4 \end{array} + \begin{array}{c} 4 \\ \vdots \\ 2 \end{array} \begin{array}{c} \nearrow \\ \text{---} \\ \searrow \end{array} \begin{array}{c} 1 \\ \vdots \\ 3 \end{array}$$

$$ig \bar{\psi}_3 \frac{-i}{(p_1 + p_2)} (ig) \psi_2 + ig \bar{\psi}_3 \frac{-i}{(p_2 + p_4)} ig \psi_2$$

$$-ig^2 \bar{\psi}_3 \frac{(p_1 + p_2)}{(p_1 + p_2)^2} \psi_2 - ig^2 \bar{\psi}_3 \frac{(p_2 + p_4)}{(p_2 + p_4)^2} \psi_2$$

Choose

$$iA_4(\phi \bar{f}_2^+ f_3^- \phi_a):$$

$$-g^2 \left(\frac{\langle 3 | 1+2 | 2 \rangle}{\langle 12 \rangle [12]} + \frac{\langle 3 | 2+4 | 2 \rangle}{\langle 24 \rangle [24]} \right)$$

Recall massless weyl

$$P = P_{\alpha\dot{\alpha}} = P_{\mu} \sigma^{\mu}_{\alpha\dot{\alpha}} \quad \text{and} \quad \text{so}$$

$$\langle P \rangle P = 0 = P | P \rangle$$

$$= -g^2 \left(\frac{\langle 3 | 1 | 2 \rangle}{\langle 12 \rangle [12]} + \frac{\langle 3 | 4 | 2 \rangle}{\langle 24 \rangle [24]} \right)$$

$$\langle 3 | \cdot - | 1 \rangle^{\dot{\alpha}} [1^{\alpha} | 2]_{\alpha} = - \langle 3 | \cdot \rangle [12]$$

$$= g^2 \left(\frac{\langle 31 \rangle [12]}{\langle 12 \rangle [12]} + \frac{\langle 34 \rangle [42]}{\langle 24 \rangle [24]} \right)$$

$$= g^2 \left(\frac{\langle 31 \rangle}{\langle 12 \rangle} - \frac{\langle 34 \rangle}{\langle 24 \rangle} \right) = -g^2 \left(\frac{\langle 13 \rangle}{\langle 12 \rangle} + \frac{\langle 43 \rangle}{\langle 42 \rangle} \right)$$

$$A_u(\phi \bar{f}^+ f^- \phi) = -g^2 \left(\frac{\langle 13 \rangle}{\langle 12 \rangle} + \frac{\langle 34 \rangle}{\langle 24 \rangle} \right)$$

* Bose symmetry w.r.t. $1 \leftrightarrow 4$ (Scalar Particles)

Momentum Conservation: $\sum_{i=1}^n p_i^\mu = 0$

$$\rightarrow \sigma_\mu \sum_{i=1}^n p_i^\mu = 0 = \sum_{i=1}^n p_{\alpha\dot{\alpha}}^i = - \sum_{i=1}^n |i\rangle [i| \Leftrightarrow$$

$$\sum_{i=1}^n |i\rangle [i| = 0 \quad \left(\begin{array}{l} \text{Spinor-helicity momenta} \\ \text{Conservation} \end{array} \right)$$

Take any light like vectors q^μ, k^μ :

$$\sum_{i=1}^n \langle q | i \rangle [i | k \rangle = 0$$

ex: $n=4$

$$|1\rangle [1| + \dots + |4\rangle [4| = 0$$

$$\left\{ \begin{array}{l} \langle 21 \rangle [14] + \langle 23 \rangle [34] = 0 \\ \langle 12 \rangle [23] + \langle 14 \rangle [43] = 0 \end{array} \right\}$$

* For all outgoing momenta 

$$S_{ij} \equiv -(p_i + p_j)^2$$

$$S_{ijk} \equiv -(p_i + p_j + p_k)^2$$

(Mandelstam Variables)

$$s = S_{12}$$

$$t = S_{13}$$

$$u = S_{14}$$

← For 4 particle process

* Calculate the Spin-sum:

$$\langle |A_u(\phi \bar{f} f \phi)|^2 \rangle = \sum |A_u(\phi, \bar{f}^{h_2} f^{h_3} \phi)|^2$$

* We already know the products $\bar{u}(1) \gamma$ vanish unless the helicity states are opposite

$$\langle |A_u(\phi \bar{f} f \phi)|^2 \rangle = \underbrace{|A_u(\phi \bar{f}^+ f^- \phi)|^2 + |A_u(\phi \bar{f}^- f^+ \phi)|^2}_{\text{Already know}}$$

$$A_u(\phi \bar{f}^- f^+ \phi) = ig \bar{u}_3 \frac{-i}{(\not{p}_1 + \not{p}_2)} ig v_2^- + [1 \leftrightarrow 4]$$

$$= -ig^2 \frac{[3|1+2|2\rangle}{\langle 12 \rangle [12]} + 1 \leftrightarrow 4$$

$$A_u(\phi \bar{f}^- f^+ \phi) = -g^2 \left(\frac{[3|1|2\rangle}{\langle 12 \rangle [12]} + \frac{[3|4|2\rangle}{\langle 42 \rangle [42]} \right)$$

$$= g^2 \left(\frac{[31] \langle 12 \rangle}{\langle 12 \rangle [12]} + \frac{[34] \langle 42 \rangle}{\langle 42 \rangle [42]} \right)$$

$$= g^2 \left(\frac{[31]}{[12]} + \frac{[34]}{[42]} \right)$$

$$A_u(\phi \bar{f}^- f^+ \phi) = -g^2 \left(\frac{[13]}{[12]} + \frac{[34]}{[24]} \right)$$

* Smarter: flipping helicity for 2 and 3 means $|2\rangle \rightarrow [2]$ and $|3\rangle \rightarrow [3]$

so we could have taken $A_u(\phi \bar{f}^+ f^- \phi)$ and simply replaced $\langle \rangle$ by $[]$

when we square this we get the exact same thing for both terms, so we focus on one

$$|A_u(\phi \bar{f} f \phi)|^2 = g^4 \left(\frac{[13]}{[12]} + \frac{[34]}{[24]} \right) \left(\frac{\langle 13 \rangle}{[12]} + \frac{\langle 34 \rangle}{[24]} \right)$$

$$= g^4 \left(\frac{\langle 13 \rangle [13]}{\langle 12 \rangle [12]} + \frac{\langle 13 \rangle [34]}{\langle 12 \rangle [24]} + \frac{[13] \langle 34 \rangle}{[12] \langle 24 \rangle} + \frac{\langle 34 \rangle [34]}{\langle 24 \rangle [24]} \right)$$

$$= g^4 \left(\frac{(p_1 + p_3)^2}{(p_1 + p_2)^2} + \frac{(p_1 + p_2)^2}{(p_1 + p_3)^2} + \frac{\langle 13 \rangle [34]}{\langle 12 \rangle [24]} + \frac{[13] \langle 34 \rangle}{[12] \langle 24 \rangle} \right)$$

$$= g^4 \left(\frac{t}{s} + \frac{s}{t} + (1) \right) \quad \text{Use momentum conservation}$$

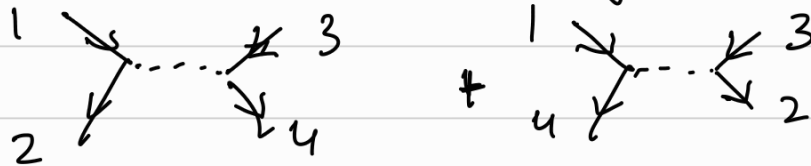
$$\left. \begin{aligned} \langle 12 \rangle [24] &= -\langle 13 \rangle [34] \\ [12] \langle 24 \rangle &= -[13] \langle 34 \rangle \end{aligned} \right\} \quad \text{Thurs}$$

$$|A_u(\phi \bar{f} f \phi)|^2 = g^4 \left(\frac{t}{s} + \frac{s}{t} - 2 \right) = g^4 \left(\frac{t^2 + s^2 - 2st}{st} \right)$$

$$\therefore \langle |A_u(\phi \bar{f} f \phi)|^2 \rangle = 2g^4 \frac{(t-s)^2}{st}$$

ex 2.6] Calculate $A_u(\bar{f}^- f^- \bar{f}^- f^-)$ in $\phi \bar{\phi} \phi$

$$\langle \Omega | \bar{\psi}_1 \psi_2 \bar{\psi}_3 \psi_4 | \Omega \rangle \sim ig^2 \langle \phi_2 \bar{\psi}_2 \psi_2 \bar{\psi}_4 \psi_4 \phi_2 \rangle$$



$$ig \langle 12 \rangle \frac{-i}{(p_1 + p_2)^2} \langle 34 \rangle ig + ig \langle 14 \rangle \frac{-i}{(p_1 + p_4)^2} \langle 32 \rangle$$

$$= ig^2 \left(\frac{\langle 12 \rangle \langle 34 \rangle}{\langle 12 \rangle [12]} + \frac{\langle 14 \rangle \langle 32 \rangle}{\langle 14 \rangle [14]} \right)$$

$$\Rightarrow A_u(\bar{f}^- f^- \bar{f}^- f^-) = g^2 \left(\frac{\langle 34 \rangle}{[12]} - \frac{\langle 23 \rangle}{[14]} \right)$$

$$\frac{\langle 34 \rangle [14] - [12] \langle 23 \rangle}{[12] [14]}$$

$$\langle 32 \rangle [21] = -\langle 34 \rangle [41]$$

$$\langle 23 \rangle \langle 12 \rangle = \langle 34 \rangle \langle 14 \rangle$$

...

I think only the S-channel contributes

$$A_u(\bar{f}^- f^- \bar{f}^- f^-) = g^2 \frac{\langle 34 \rangle}{\langle 12 \rangle}$$

$$|A_u(\bar{f}^- f^- \bar{f}^- f^-)|^2 = g^4 \frac{\langle 34 \rangle \langle 34 \rangle}{\langle 12 \rangle \langle 12 \rangle} = \frac{(p_3 + p_4)^2}{(p_1 + p_2)^2} g^4$$

$$\therefore |A_u(\bar{f}^- f^- \bar{f}^- f^-)|^2 = g^4$$

$$\langle |A_u(\bar{f} f \bar{f} f)|^2 \rangle = 6 g^4$$

→ Not sure of this value as we have
 $(--++) + (-++-) + (+--+) + (----) + (++++)$
 $+ (++--)$

~> Yup, I think these are all of them ...

$$\text{Note: } A_u(\bar{f}^- f^- \bar{f}^+ f^+) = g^2 \frac{\langle 12 \rangle}{\langle 34 \rangle} = g^2 \frac{[34]}{[12]}$$

From this we can deduce all other factors

$$A_u(\bar{f}^+ f^+ \bar{f}^- f^-) \quad \begin{array}{l} \text{relabel} \\ 1 \leftrightarrow 3 \\ 2 \leftrightarrow 4 \end{array}$$

proof:

$$i g^2 \frac{1}{\langle 43 \rangle (p_1 + p_2)^2} [21] \quad \langle 34 \rangle$$

$$= i g^2 \langle 34 \rangle [12] / \langle 12 \rangle [12] = i g^2 \frac{\langle 34 \rangle}{\langle 12 \rangle}$$

• Likewise all the others are permutations
 therefore: ~~6~~ is correct.

ex 2.8

$$\mathcal{L} = i \bar{\psi} \not{\partial} \psi - \partial_\mu \bar{\psi} \partial^\mu \psi + \frac{1}{2} g \bar{\psi} \psi \psi \psi$$

$$+ \frac{1}{2} g \bar{\psi} \psi \psi \psi - \frac{\lambda}{4} |\phi|^4$$

Show

$$(a) A_u(\phi\phi\bar{\phi}\bar{\phi}) = -\lambda$$

Proof: all terms from $\phi\phi\phi$ yield tadpoles and so the only contribution comes from $\lambda|\phi|^4$.

$$\langle \Omega | \phi\phi\bar{\phi}\bar{\phi} | \Omega \rangle_c = -\frac{i}{4}\lambda \int d^4z \Delta_{12} \Delta_{23} \dots$$

$$\langle \Omega | \phi\phi\bar{\phi}\bar{\phi} | \Omega \rangle_c = 4 \left(\text{diagram with cross} \right) = -i\lambda$$

$$\therefore A_u(\phi\phi\bar{\phi}\bar{\phi}) = -\lambda$$

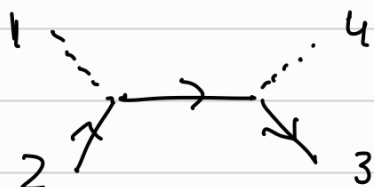
• Do not consider loops yet

$$(b) A_u(\phi\phi^* \phi^* \bar{\phi}) = -|g|^2 \langle 24 \rangle \langle 34 \rangle$$

$$\langle \Omega | \phi\phi\phi^*\bar{\phi} | \Omega \rangle_c$$

$$O(\lambda^0) + \langle 0 | \phi\phi\phi^*\bar{\phi} \int d^4z d^4w \left(\frac{1}{2}g\phi\phi + \frac{1}{2}g^*\phi^*\phi^* \right)^2$$

$$+ 2 \int d^4z d^4w \langle 0 | \phi\phi\phi^*\bar{\phi} \overbrace{g g^* \phi_z \phi_z^* \phi_w^* \phi_w \bar{\phi}}^{\text{diagram}} | 0 \rangle$$



Note since 1 and 4 are not identical particles there is no exchange symmetry

$$i A_u(\phi\phi^* \phi^* \bar{\phi}) = i g \bar{v}_3^+ \frac{-i(\not{p}_1 + \not{p}_2)}{(p_1 + p_2)^2} v_2^- i g^*$$

$$= i |g|^2 \frac{[3|1+2|2\rangle}{\langle 12 \rangle [12]} = -i |g|^2 \frac{[31] \langle 12 \rangle}{\langle 12 \rangle [12]}$$

$$\therefore A_u(\phi\phi^* \phi^* \bar{\phi}) = -|g|^2 [31]/[12]$$

Since

$$\langle 42 \rangle [21] + \langle 43 \rangle [31] = 0$$

$$\langle 24 \rangle [12] = [31] \langle 34 \rangle$$



$$\Rightarrow [31]/[12] = \langle 24 \rangle / \langle 34 \rangle$$

$$A_4(\phi \psi \psi^\dagger \bar{\phi}) = -|g|^2 \frac{\langle 24 \rangle}{\langle 34 \rangle}$$

$$\textcircled{c} A_4(f^\dagger f^\dagger f^\dagger f^\dagger):$$

again we'll have the above, except with the middle prop. given by $\phi \cdots \cdots \bar{\phi}$

$$ig^* \bar{u}_4 \gamma_3^+ \frac{-i}{(p_1 p_2)^2} \bar{u}_2 \gamma_1^- ig$$

$$\frac{[43] \langle 21 \rangle}{\langle 12 \rangle [12]} i |g|^2 = \frac{[34] i |g|^2}{[12]}$$

$$\langle 12 \rangle [12] = \langle 34 \rangle [34] \Rightarrow \frac{[34]}{[12]} = \frac{\langle 12 \rangle}{\langle 34 \rangle}$$

$$A_4(f^\dagger f^\dagger f^\dagger f^\dagger) = |g|^2 \frac{\langle 12 \rangle}{\langle 34 \rangle}$$

Schouten identity:

* Given 3 vectors on a plane, they can not all be linearly independent

$\rightarrow |i\rangle$ are all two component spinors
 $\alpha, \beta = 1, 2$

$\rightarrow$ So, given three 2-component spinors
 $|i\rangle, |j\rangle, |k\rangle$

$$|k\rangle = a|i\rangle + b|j\rangle$$

$$\langle i|k\rangle = b\langle ij\rangle$$

$$\langle j|k\rangle = a\langle ji\rangle$$

$$\frac{\langle i|k\rangle}{\langle ij\rangle}$$

$$b = \frac{\langle i|k\rangle}{\langle ij\rangle}$$

$$a = \langle j|k\rangle / \langle ji\rangle$$

$$|k\rangle = (|i\rangle \langle j|k\rangle) / \langle ji\rangle + \left(\frac{\langle i|k\rangle}{\langle ij\rangle} |j\rangle \right)$$

$$\langle ij\rangle |k\rangle = -|i\rangle \langle j|k\rangle - \langle k|j\rangle |i\rangle$$



$$|i\rangle\langle jk\rangle + |j\rangle\langle ki\rangle + |k\rangle\langle ij\rangle = 0$$

(Schouten identity)



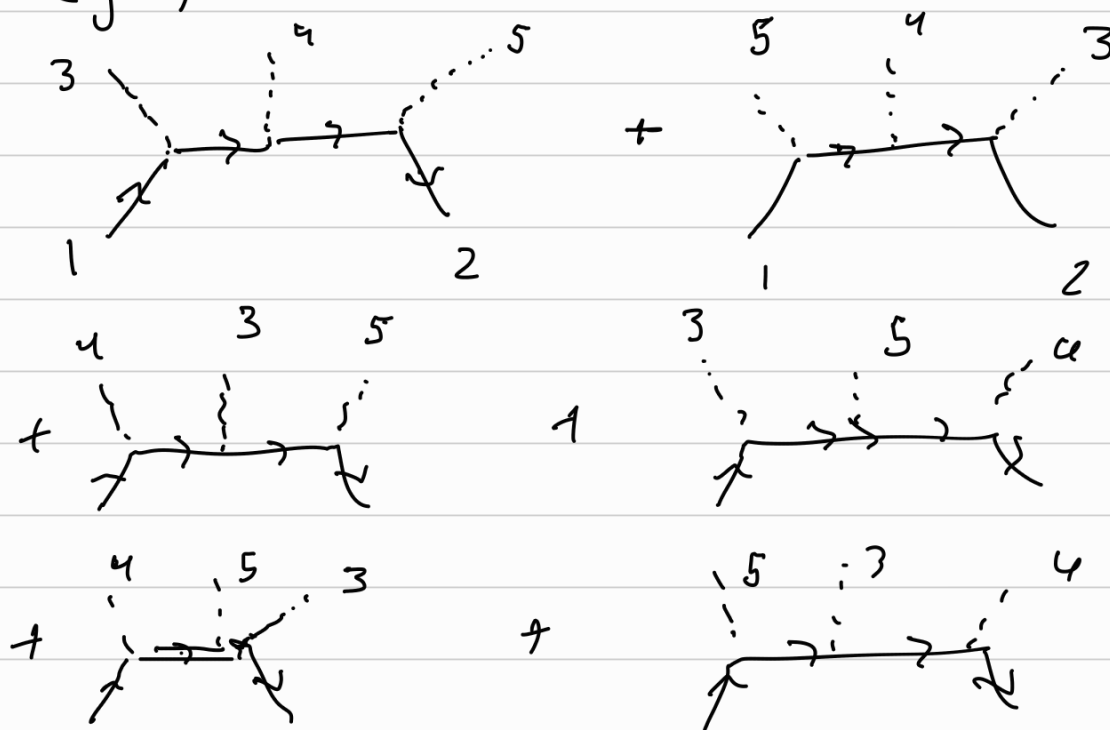
$$|i\rangle\langle jk\rangle + |j\rangle\langle ki\rangle + |k\rangle\langle ij\rangle = 0$$

ex 2.9 | $A_5(f^{-}\bar{f}^{-}\phi\phi\phi)$ in $g\phi\psi\psi$

$\mathcal{O}(g^1)$: Bubbles / sub diagrams (Bubble) $\times A_3$

$$\mathcal{O}(g^2) = 0$$

$\mathcal{O}(g^3)$:



$$\frac{1}{3!} (\phi_2 \bar{\phi}_2 \phi_2 \phi_\omega \bar{\phi}_\omega \phi_\omega \phi_2 \bar{\phi}_2 \phi_2) \psi \bar{\psi} \phi_3 \phi_4 \phi_5$$

$$= \underbrace{f \bar{f} \phi_3 \phi_4 \phi_5}_{\text{diagram 1}} + \underbrace{f \bar{f} \phi \phi \phi}_{\text{diagram 2}} + \underbrace{f \bar{f} \phi \phi \phi}_{\text{diagram 3}} + \underbrace{f \bar{f} \phi \phi \phi}_{\text{diagram 4}}$$

$$A_5(f^{-}\bar{f}^{-}\phi\phi\phi) : \text{diagram 1}$$

only way to make sense of

$$\tilde{u} = (P_1) = \langle P_1 | \tilde{u}$$

$$v = (P_2) = | 2 \rangle \tilde{v}$$

$$= (ig) \langle 1 | (ig) \left(\frac{-i (2+3+4)}{(P_2+P_3+P_4)^2} \right) \left(\frac{-i (2+3)}{(P_2+P_3)^2} \right) (ig | 2 \rangle)$$

$$= i^2 i^2 i g^3 \langle 1 | \frac{(-1-5)(2+3)}{(P_1+P_5)^2 (P_2+P_3)^2} | 2 \rangle$$

$$= -ig^3 \frac{\langle 1 | 5 3 | 2 \rangle}{(P_1+P_5)^2 (P_2+P_3)^2} \quad 53 = -P_5 P_3 = \frac{1}{2} \langle 35 \rangle [35]$$

$$= \frac{ig^3}{2} \frac{\langle 35 \rangle [35] \langle 12 \rangle}{\langle 15 \rangle [15] \langle 23 \rangle [23]} + \dots$$

not too sure

Massless vectors and examples from QED

• For spin 1: "dot" in their polarization vector

$$\epsilon_-^\mu(p; q) \equiv - \frac{\langle q | \gamma^\mu | p \rangle}{\sqrt{2} [qP]}, \quad \epsilon_+^\mu(p; q) \equiv - \frac{\langle q | \gamma^\mu | p \rangle}{\sqrt{2} \langle qP \rangle}$$

(Polarization Vectors)

• $P \neq q$

(Claim: $P_\mu \epsilon_\pm^\mu(p) = 0$)

$$\text{Proof: } + : P_\mu \left(- \frac{\langle q | \gamma^\mu | p \rangle}{\sqrt{2} \langle qP \rangle} \right) = - \frac{\langle q | \gamma^\mu | p \rangle P_\mu}{\sqrt{2} \langle qP \rangle} = - \frac{\langle q | \gamma^\mu P_\mu | p \rangle}{\sqrt{2} \langle qP \rangle} = - \frac{\langle q | \not{P} | p \rangle}{\sqrt{2} \langle qP \rangle}$$

$P^\mu P_\mu | p \rangle = 0$ (Massless Weyl)

$$- : P_\mu \left(- \frac{\langle q | \gamma^\mu | p \rangle}{\sqrt{2} [qP]} \right) = - \frac{\langle q | \gamma^\mu | p \rangle P_\mu}{\sqrt{2} [qP]} = - \frac{\langle q | \gamma^\mu P_\mu | p \rangle}{\sqrt{2} [qP]} = 0 \quad \checkmark$$

$$\sigma_\mu^\alpha \epsilon_+^\mu(p, q) = - \frac{\langle q | \sigma_\mu \sigma^\mu | p \rangle}{\sqrt{2} \langle qP \rangle} = \tilde{q}$$

$$= \tilde{q} (\sigma_\mu)_{\alpha\dot{\alpha}} \langle q | \bar{\sigma}^{\mu\dot{\beta}\beta} | p \rangle_\beta$$

$$= \tilde{q} \epsilon^{\dot{\beta}\gamma} \epsilon^{\beta\gamma} ((\sigma^\mu)_{\alpha\dot{\alpha}} (\sigma_\mu)_{\gamma\dot{\gamma}}) \langle q | \bar{\sigma}^{\mu\dot{\beta}\beta} | p \rangle_\beta$$

$$= \tilde{q} \epsilon^{\dot{\beta}\gamma} \epsilon^{\beta\gamma} (-2 \epsilon_{\alpha\gamma} \epsilon_{\dot{\alpha}\dot{\gamma}}) \langle q | \bar{\sigma}^{\mu\dot{\beta}\beta} | p \rangle_\beta$$

$$= (-2\tilde{q}) \delta_{\alpha\dot{\alpha}} \delta_{\beta\dot{\beta}} \langle q | \bar{\sigma}^{\mu\dot{\beta}\beta} | p \rangle_\beta$$

$$= \sqrt{2} \dots$$

$$= \frac{1}{2\sqrt{q}} \langle p | \sigma_\mu | q \rangle$$

which is close, should actually be

$$\chi_\mu \in \mathbb{H}_\pm(p, q) \text{ so}$$

$$= \begin{pmatrix} 0 & \sigma_\mu \in \mathbb{H}_\pm \\ \bar{\sigma}_\mu \in \mathbb{H}_\pm & 0 \end{pmatrix} = |p\rangle [q| + |q\rangle \langle p|$$

This notation is really

$$\begin{pmatrix} 0 & |p\rangle \langle q| \\ |q\rangle \langle p| & 0 \end{pmatrix} \quad \text{Ⓢ}$$

$\therefore$

$$\not\epsilon_-(p, q) = \frac{\sqrt{2}}{\langle q p \rangle} (|p\rangle [q| + |q\rangle \langle p|)$$

$$\not\epsilon_+(p, q) = \frac{\sqrt{2}}{\langle q p \rangle} (|p\rangle \langle q| + |p\rangle \langle q|)$$

* The arbitrariness of q reflects the gauge invariance

$\rightarrow$ One is free to shift the polarization vectors

$$\epsilon_\pm^\mu \rightarrow \epsilon_\pm^\mu + c p^\mu$$

$$\text{Since } \partial_\mu A^\mu = 0 \text{ (Ward identity)}$$

$$\text{QED: } \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + i \bar{\Psi} \gamma^\mu (\partial_\mu - i e A_\mu) \Psi$$

interaction term $e \bar{\Psi} \gamma^\mu A_\mu \Psi$

Vertex Rule: $(i e \gamma^\mu)$

• Each photon: ϵ^μ , $h = \pm 1$

$$\text{ex: } A_3(f^{h_1} \bar{f}^{h_2} \gamma^{h_3}) \quad ; \quad f = e^-, \quad \bar{f} = e^+ \text{ (outgoing)}$$

$$h_1 = -1/2, \quad h_2 = 1/2, \quad h_3 = -1$$

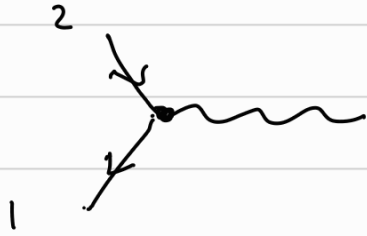
$$\therefore i A_3(f^- \bar{f}^+ \gamma^-) = \bar{u}_-(p_1) i e \gamma_\mu u_+(p_2) \epsilon_-^\mu(p_3; q)$$

$$= i e \langle 2 | \gamma_\mu | 1 \rangle \epsilon_-^\mu(p_3; q)$$

could also write this as

$$\langle 2 | \gamma_\mu | 1 \rangle$$

$$= \langle q | \gamma^\mu | 3 \rangle$$



$$iA_3(f^+ \bar{f}^+ \gamma^-) = ie \langle 11 | \gamma_\mu | 2 \rangle \left(-\frac{1}{\sqrt{2}} \frac{[q3]}{[3q]} \right)$$

$$= (ie/\sqrt{2}) \langle 11 | \gamma_\mu | 2 \rangle \langle 3 | \gamma^\mu | q \rangle \frac{1}{[3q]}$$

$$= \sqrt{2} ie \langle 13 \rangle [2q]$$

$$A_3(f^+ \bar{f}^+ \gamma^-) = \tilde{e} \langle 13 \rangle [2q] / [3q] \quad \underline{\tilde{e} \equiv \sqrt{2} ie}$$

We now see how to get rid of the reference

Spinor q

$$A_3 = \tilde{e} \frac{\langle 13 \rangle [2q]}{[3q]}$$

Multiply By factor of 1 which has momenta in $[pq]$ and elsewhere without q . Then, mult by the other:

$$\frac{\langle 13 \rangle [2q]}{[3q]}$$

$$\langle 12 \rangle$$

Thus, we multiply by $\langle 12 \rangle$

$$\frac{\langle 13 \rangle \langle 12 \rangle [2q]}{\langle 12 \rangle [3q]}, \quad p_2 = -p_1 - p_3$$

$$\rightarrow \langle 12 \rangle [2q] = -\langle 11 | 2 | q \rangle = \langle 1 | 1+3 | q \rangle = \langle 1 | 3 | q \rangle$$

$$= -\langle 13 \rangle [3q]$$

$$A_3 = \tilde{e} \frac{-\langle 13 \rangle^2}{\langle 12 \rangle}$$

$\rightarrow A_3$ only depends on $\langle \rangle$ brackets which is a consequence of 3-particle special kinematics if 3 light-like vectors satisfy

$$\underline{p_1^\mu + p_2^\mu + p_3^\mu = 0}$$

then

$$\langle 12 \rangle [12] = 2 p_1 \cdot p_2 = (p_1 + p_2)^2 = p_3^2 = 0$$

$$\text{So either } \underline{\langle 12 \rangle = 0} \quad \text{or} \quad \underline{[12] = 0}$$

i.e. Suppose $\langle 12 \rangle \neq 0$, then

$$\langle 1 | (11) [11] + 12 \rangle [21] + 13 \rangle [31] = 0 \quad) \quad 13]$$

$$\langle 12 \rangle [23] = 0$$

$$\bullet \text{ So } [23] = 0$$

$$\langle 21 \rangle [13] = 0$$

$$[12] = 0, [23] = 0$$

$$[13] = 0$$

• This implies that the spinors are proportional

$$|1\rangle \propto |2\rangle \propto |3\rangle$$

(ii) Assuming $[12] \neq 0$

$$|1\rangle \propto |2\rangle \propto |3\rangle$$

* From this we conclude:

(i) A non-vanishing on-shell 3-particle amp. w only massless particles can only depend on $\langle \rangle$ or $[\]$

(ii) On shell 3-particle amp. of only massless particles can only be non-vanishing in complex momenta.

* Finally, note $|1\rangle \propto |2\rangle$ causes $\epsilon_{-}^{n}(3;4)$ to be divergent so it is not "allowed"

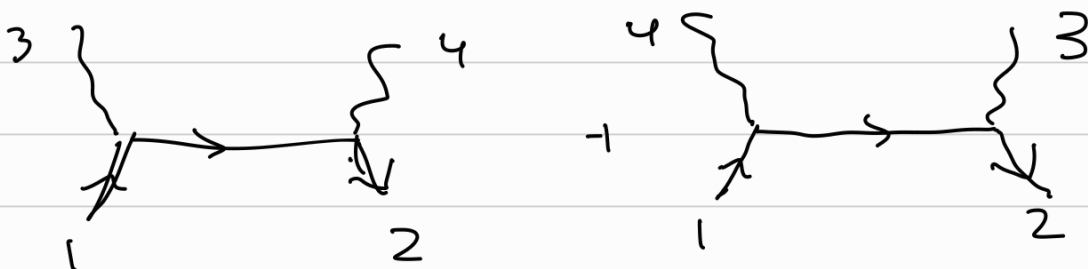
ex: $e^- \gamma \rightarrow e^- \gamma$

$$A_4(\bar{f} f \gamma \gamma)$$

$$\langle \Omega | \bar{\psi} \psi A^{\mu} A^{\nu} | \Omega \rangle$$

$$e \bar{\psi} \not{A} \psi$$

(Compton-Scattering)



$$\bar{u}_2 \not{\epsilon}_4 \frac{-i(\not{p}_1 + \not{p}_3)}{(\not{p}_1 + \not{p}_3)^2} \not{\epsilon}_3 v_1 (ie)^2 + (ie)^2 \bar{u}_2 \not{\epsilon}_3 \frac{-i(\not{p}_1 + \not{p}_4)}{(\not{p}_1 + \not{p}_4)^2} \not{\epsilon}_4 v_1$$

• We have $\bar{u}_2 (\gamma^{\mu} \gamma^{\nu} \gamma^0) v_1$
however

$$(\not{0} \not{\sigma}^{\mu}) (\not{2} \not{\sigma}^{\nu}) (\not{0} \not{\sigma}^0) = (\not{0} \not{\sigma}^{\mu}) (\not{2} \not{\sigma}^{\nu} \not{0}) = 0$$

$$\begin{aligned}
 & \begin{pmatrix} \bar{u}^m 0 \\ 0 \end{pmatrix} \begin{pmatrix} \bar{u}^n 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ \bar{u}^o 0 \end{pmatrix} = \begin{pmatrix} \bar{u}^m 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & \bar{u}^n \sigma^o \end{pmatrix} \\
 & = \begin{pmatrix} 0 & \sigma^m \bar{u}^n \sigma^o \\ \bar{u}^m \sigma^n \bar{u}^o & 0 \end{pmatrix} \\
 & \text{so } (\bar{u}_1, \bar{u}_2) \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\
 & = (\bar{u}_1, \bar{u}_2) \begin{pmatrix} \sigma v_2 \\ \sigma v_1 \end{pmatrix} = \bar{u}_1 \sigma v_2 + \bar{u}_2 \sigma v_1
 \end{aligned}$$

* ODD γ 's : u and v must have opposite helicity

$$\begin{aligned}
 & * \text{EVEN } \gamma\text{'s} : (\bar{u}_1, \bar{u}_2) \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\
 & = (\bar{u}_1, \bar{u}_2) \begin{pmatrix} \sigma v_1 \\ \sigma v_2 \end{pmatrix} = \dots
 \end{aligned}$$

Must have the same

$\overbrace{\gamma^\mu \dots \gamma^\sigma}^{\text{Odd}}$	$\longleftrightarrow$	$(\bar{u}_\pm, v_\mp)$
$\overbrace{\gamma^\mu \dots \gamma^\sigma}^{\text{Even}}$	$\longleftrightarrow$	$(\bar{u}_\pm, v_\pm)$

(1) Suppose Both photons have neg. hel.
 $h_\gamma = -1$:

↖ (choosing opposite, doesn't matter)

$$(ie)^2 \langle 2 | (|4\rangle [q] + |q\rangle \langle 4 |) (\quad) (|3\rangle [q_3] + |q_3\rangle [3]) |1\rangle$$

$$\hookrightarrow |3\rangle [q_3, 1]$$

Choose $[q_3]$ & $[1]$ which makes the amplitude zero.

(2) Choose $|4\rangle$ & $|1\rangle$ so that the

Second diagram variables as well

$$\therefore \underline{A_u(\bar{f} + f - \gamma - \gamma^-)} = 0$$

* Proof of the above without a ref. to q_3
 $A_u(\bar{f} + f - \gamma - \gamma^-)$

$$= \frac{2ie^2 \langle 24 \rangle \langle 3 | 1+3 | q_u \rangle [q_u 1]}{[q_u 4] \langle 13 \rangle [13] [q_3 3]} + \frac{2ie^2 \langle 23 \rangle \langle 4 | 1+4 | q_3 \rangle [q_u 1]}{[q_3 3] \langle 14 \rangle [14] [q_4 4]} \quad \text{Used}$$

$$\downarrow \quad \frac{\langle 24 \rangle}{[13]} = \frac{\langle 23 \rangle}{[41]}$$

$$= \frac{2ie^2 \langle 23 \rangle}{[14] [q_3 3] [q_4 4]} \left(- \frac{\langle 3 | 1 | q_u \rangle [q_u 1]}{\langle 13 \rangle} + \frac{\langle 4 | 1 | q_3 \rangle [q_u 1]}{\langle 14 \rangle} \right)$$

$$= \frac{2ie^2 \langle 23 \rangle}{\langle 14 \rangle \langle 13 \rangle [14] [q_3 3] [q_4 4]} \left(\langle 13 \rangle \langle 4 | 1 | q_3 \rangle [q_u 1] - \langle 14 \rangle \langle 3 | 1 | q_u \rangle [q_u 1] \right)$$

$$= 0 \quad \checkmark$$

* Let us now consider $A_u(\bar{f} + f - \gamma + \gamma^-)$

$$A_u = \frac{2e^2 \langle 2 | 4 \rangle [q_u | (-1) \langle 11 - 13 \rangle \langle 31 | q_3 \rangle [31]}{[q_u 4] \langle 13 \rangle [13] \langle q_3 3 \rangle}$$

$$+ \frac{2e^2 \langle 2 q_3 \rangle [3 | (-1) \langle 11 - 1 q_u \rangle \langle 41 | q_4 \rangle [q_u 1]}{[q_u 4] \langle 14 \rangle [14] \langle q_3 3 \rangle}$$

$$= - \frac{2e^2 \langle 24 \rangle ([q_u 1] \langle 14 \rangle + [q_u 3] \langle 3 q_3 \rangle) [31]}{[q_u 4] \langle 13 \rangle [13] \langle q_3 3 \rangle}$$

$$= \frac{2e^2 \langle 2 q_3 \rangle [31] [q_u 1]}{\langle 14 \rangle [14] [q_u 4] \langle q_3 3 \rangle}$$

$$\rightarrow q_3 = q_4 = p_1, \text{ then } \rightarrow 0$$

$$-2e^2 \langle 24 \rangle [13] \langle 31 \rangle [31]$$

$$A_u(\bar{f}^+ \bar{f}^- \gamma^+ \gamma^-) = \frac{[14] \langle 13 \rangle [13] \langle 13 \rangle}{[14] \langle 13 \rangle}$$

$$-2e^2 \langle 24 \rangle [13]$$

$$= \frac{[14] \langle 13 \rangle}{[14] \langle 13 \rangle} \Rightarrow \frac{[13]}{[14]} = - \frac{\langle 24 \rangle}{\langle 23 \rangle}$$

$$A_u(\bar{f}^+ f^- \gamma^+ \gamma^-) = \frac{\tilde{e}^2 \langle 24 \rangle^2}{\langle 13 \rangle \langle 23 \rangle}$$

we obtain $A_u(\bar{f}^+ f^- \gamma^- \gamma^+)$ we take $3 \leftrightarrow 4$

$$A_u(\bar{f}^+ f^- \gamma^- \gamma^+) = \tilde{e}^2 \frac{\langle 23 \rangle^2}{\langle 14 \rangle \langle 24 \rangle}$$

Proof (without reference spinors):

$$- \tilde{e}^2 \langle 24 \rangle ([q_4 1] \langle 1 q_3 \rangle + [q_4 3] \langle 3 q_2 \rangle) [3 1]$$

$$A_u = \frac{\langle 13 \rangle [13] \langle q_2 3 \rangle [q_4 4]}{\langle 14 \rangle [14] \langle q_3 3 \rangle [q_4 4]}$$

$$- \frac{\tilde{e}^2 \langle 2 q_3 \rangle [3 1] \langle 14 \rangle [q_4 1]}{\langle 14 \rangle [14] \langle q_3 3 \rangle [q_4 4]}$$

$$= -e^2 \left(\frac{\langle 24 \rangle [q_4 1] \langle 1 q_3 \rangle}{\langle 13 \rangle} - \frac{\langle 24 \rangle [q_4 3] \langle 3 q_2 \rangle}{\langle 13 \rangle} \right)$$

$$+ \frac{\langle 2 q_3 \rangle [3 1] [q_4 1]}{[14]}$$

$$\langle 23 \rangle [3 1] = - \langle 24 \rangle [4 1] \Rightarrow \frac{[3 1]}{[14]} = \frac{\langle 24 \rangle}{\langle 23 \rangle}$$

$$= \frac{e^2 \langle 24 \rangle}{\langle q_3 3 \rangle [q_4 4] \langle 13 \rangle \langle 23 \rangle} \left(\begin{aligned} & \langle 23 \rangle [q_4 1] \langle 1 q_3 \rangle \\ & - \langle 23 \rangle [q_4 3] \langle 3 q_2 \rangle \\ & + \langle 2 q_3 \rangle \langle 13 \rangle [q_4 1] \end{aligned} \right)$$

$$(middle) = - \langle 23 \rangle [3 q_4] = \langle 2 | 3 | q_4 \rangle$$

$$= - \langle 2 | (1 + 4) | q_4 \rangle = \langle 21 \rangle [1 q_4] + \langle 24 \rangle [4 q_4]$$

$$\langle 23 \rangle [q_4 1] \langle 1 q_3 \rangle + \langle 21 \rangle [1 q_4] \langle 3 q_3 \rangle$$

$$1) = + \langle 2u \rangle [4q_u] \langle 3q_3 \rangle - \langle 2q_3 \rangle \langle 13 \rangle [4u1]$$

(y term we want so all others cancel

$$\langle 23 \rangle [4u1] \langle 1q_3 \rangle + \langle 21 \rangle [1q_u] \langle 3q_3 \rangle + \langle 2q_3 \rangle \langle 13 \rangle [4u1] \\ [4u1] (\langle 23 \rangle \langle 1q_3 \rangle + \langle 21 \rangle \langle 3q_3 \rangle + \langle 13 \rangle \langle 2q_3 \rangle)$$

is this a shortcut?

$$12 \rangle \langle 23 \rangle + 12 \rangle \langle 31 \rangle + 13 \rangle \langle 12 \rangle = 0$$

$$\langle q_3 1 \rangle \langle 23 \rangle + \langle q_3 2 \rangle \langle 31 \rangle + \langle q_3 3 \rangle \langle 12 \rangle = 0$$

$$+ \langle 23 \rangle \langle 1q_3 \rangle + \langle 21 \rangle \langle 3q_3 \rangle + \langle 13 \rangle \langle 2q_3 \rangle$$

off by a (-) sign but that's my fault $\square$

Scalar QED: $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - |D\phi|^2 - \frac{1}{6} \mu^2 \phi^2$

Feynman Rules:

$i i | \phi \phi^* A^\mu$: vertex: $i e (p_2 - p_1)^\mu$

$i i | \phi \phi^* A^\mu A_\mu$: vertex: $-2 i e^2 \eta_{\mu\nu}$

$i i | \phi |^4$: vertex: $-i \lambda$

(ψ, ψ^*) : Supersymmetric spin-0 superpartner of (e^+, e^-)

$\psi + \gamma \rightarrow \psi + \gamma$ (Spin 0 Compton)

Take high energy limit $m \rightarrow 0$ & m

ex 2.15) $\mathcal{A}_3(\psi \psi^* \gamma^-)$

only $i e A^\mu$ ()

$-2 p_1 p_2 = -\langle 12 \rangle \langle 3 \rangle$

$i \mathcal{A}_3 = i e E^\mu (p_2 - p_1)_\mu$

$= \frac{-i e}{\sqrt{2} [q_3 3]} (\langle q_3 | 2 | 3 \rangle - \langle q_3 | 1 | 3 \rangle)$

$= \frac{-i e}{\sqrt{2} [q_3 3]} \langle q_3 | 2 | 3 \rangle = \frac{i e}{\sqrt{2} [q_3 3]} \langle 3 2 \rangle [2 q_3] \langle 12 \rangle$

$$\sqrt{2} \langle 12 \rangle [23] = - \langle 13 \rangle [32]$$

$$A_3(\phi\phi^*\gamma^-) = \frac{e \langle 32 \rangle \langle 13 \rangle [423]}{\sqrt{2} \langle 12 \rangle} = \frac{e \langle 32 \rangle \langle 13 \rangle}{\sqrt{2} \langle 12 \rangle}$$

$$A_2(\phi\phi^*\gamma^-) = \frac{e \langle 13 \rangle \langle 32 \rangle}{\langle 12 \rangle}$$

$$A_3(\phi\phi^*\gamma^+) = - \frac{e [13] [32]}{[12]}$$

ex: 2.16 | $A_4(\phi\phi^*\gamma\gamma)$

 : comes from $e^2 A^\mu A_\mu \phi\phi^*$ interaction term

 : $(ie A^\mu (\phi \overleftrightarrow{\partial}_\mu \phi^*))^2$
is not $\#PI$ so this is the product of $A_3 A_3$

$$iA_4 = (E_{\pm 3}^+ - 2i\eta_{\mu\nu} E_{\pm 4}^\mu)$$

$$A_4 = \frac{[q_3 | \gamma^\mu | 3 \rangle [q_4 | \gamma_\mu | 4 \rangle}{[q_3 3] [q_4 4]}$$

$$= \frac{2 \langle 34 \rangle [q_3 q_4]}{[q_3 3] [q_4 4]} \quad (\text{Same helicity})$$

• Opposite helicity

$$A_4 = \frac{[q_3 | \gamma^\mu | 3 \rangle [4 | \gamma_\mu | q_4 \rangle}{[q_3 3] [q_4 4]} = \frac{2 \langle 3 q_4 \rangle [q_3 4]}{[q_3 3] [q_4 4]}$$

• For same, $q_3 = q_4 = q_{1,2}$
which vanishes

• For opposite $q_3 = 4$, $q_4 = 3$
so

A_4 contact $\rightarrow 0$

* only left with

$$\begin{array}{c} 2 \\ \{ \end{array} \begin{array}{c} a \\ \vdots \\ 1 \end{array} \begin{array}{c} a' \\ \vdots \\ 3 \end{array} \begin{array}{c} 4 \\ \} \end{array} = A_3 \frac{1}{(p_1 + p_2)^2} A_3$$

$$A(\phi \phi^* \gamma^-) = -e \frac{\langle 13 \rangle \langle 23 \rangle}{\langle 12 \rangle} \quad A(\phi \phi^* \gamma^+) = e \frac{[13][23]}{[12]}$$

Claim

$$A_4(\phi \phi^* \gamma^+ \gamma^-) = e^2 \frac{\langle 14 \rangle \langle 24 \rangle}{\langle 13 \rangle \langle 23 \rangle}$$

Proof: using

$$-e \frac{[13][43]}{[14]} \frac{1}{\langle 12 \rangle [12]} \frac{\langle a \ 4 \rangle \langle 24 \rangle}{\langle a \ 2 \rangle}$$

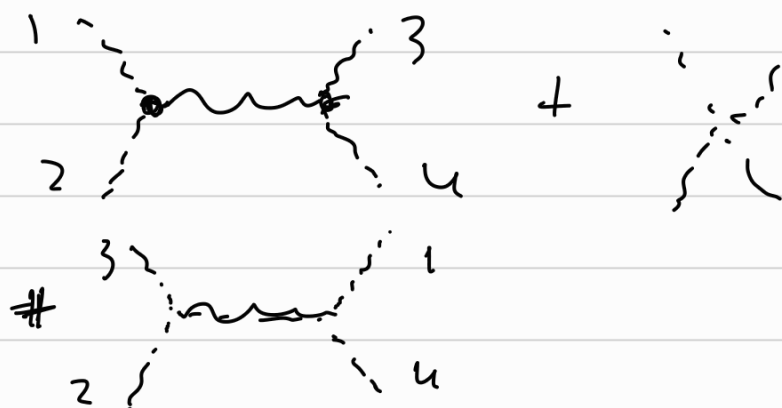
$$\cdot \langle 24 \rangle [4 \ 1] = -\langle 2 \ 1 \ 4 \ 1 \rangle$$

$$-\langle 4 \ 4 \rangle [a \ 3] = \langle 4 \ 1 \ 4 \ 3 \rangle$$

$$= \left(\frac{e [13] \langle 24 \rangle}{\langle 12 \rangle [12]} \right) \left(\frac{\langle 4 \ 1 \ 4 \ 3 \rangle}{\langle 2 \ 1 \ 4 \ 1 \rangle} \right)$$

Not sure.... Chp 3 will make this precise...

ex 2.10 $A_4(\phi \phi^* \phi \phi^*) \neq$ from $A|\phi|^4$
other from $i e A^{\mu}(\phi \overleftrightarrow{\partial}_{\mu} \phi^*)$



$$i e \int d^4x \phi^2 \partial^2 \phi^2 = i e \int d^4x \phi^2 \Box \phi^2$$

$$(ie)^2 \langle \Omega | \phi \phi \phi \phi^* | \Omega \rangle = A^\mu \phi \partial_\mu \phi^* \cdot |\Omega\rangle$$

$$= \frac{(ie(p_2 - p_1))^\mu \eta_{\mu\nu} (ie(p_3 - p_4))^\nu (-i)}{(p_1 + p_2)^2} + (1 \leftrightarrow 3)$$

$$ie^2 \frac{(p_2 p_4 - p_1 p_4 - p_2 p_3 + p_1 p_3)}{\langle 12 \rangle \langle 12 \rangle}$$

$$= ie^2 (2 p_2 p_4 - 2 p_1 p_4) / \langle 12 \rangle \langle 12 \rangle$$

$$= ie^2 (\langle 24 \rangle \langle 24 \rangle - \langle 14 \rangle \langle 14 \rangle) / \langle 12 \rangle \langle 12 \rangle$$

$$= - \frac{\langle 24 \rangle \langle 13 \rangle}{\langle 12 \rangle \langle 34 \rangle} - \frac{\langle 14 \rangle \langle 23 \rangle}{\langle 12 \rangle \langle 34 \rangle} + (1 \leftrightarrow 3)$$

$$= \frac{-1}{\langle 12 \rangle \langle 34 \rangle} (\langle 13 \rangle \langle 24 \rangle + \langle 14 \rangle \langle 23 \rangle)$$

Shouten

$$\langle 41 \rangle \langle 23 \rangle + \langle 42 \rangle \langle 31 \rangle + \langle 43 \rangle \langle 12 \rangle = 0$$

$$= \langle 14 \rangle \langle 23 \rangle + \langle 13 \rangle \langle 24 \rangle - \langle 12 \rangle \langle 34 \rangle$$

$$= \frac{(-1)}{\langle 12 \rangle \langle 34 \rangle} (2 \langle 13 \rangle \langle 24 \rangle - \langle 12 \rangle \langle 34 \rangle) + (1 \leftrightarrow 3)$$

this yields

$$A_4(\phi \phi^* \phi \phi^*) = -\lambda + e^2 \left(1 + \frac{\langle 24 \rangle^2 \langle 13 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

$$\tilde{e}^2 = 2e^2$$

+ $-\lambda$ from 141^4 term

Yang Mills, QCD, color ordering: $\mathcal{L} = -\frac{1}{4} \text{Tr}(F^{\mu\nu} F_{\mu\nu})$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - \frac{ig}{f^2} [A_\mu, A_\nu], \quad A_\mu = A_\mu^a T^a$$

$\mathcal{L} = \text{Tr}(\frac{1}{2} F_{\mu\nu} F^{\mu\nu})$ for QCD but not for

* $G = SU(N)$ General N -colors

Gluon fields in the adjoint rep $(T^a)_{bc} = -if_{abc}$

$$\dim(\square) = \dim(SU(N))$$

$$N-1 \left\{ \begin{array}{|c|c|} \hline N & N+1 \\ \hline N-1 & \\ \hline \vdots & \\ \hline 1 & \\ \hline \end{array} \right\} = 1$$

$$\dim(\square) = \frac{(N+1)N(N-1)\dots 1}{(N-2)(N-3)\dots 1} = (N+1)(N-1) = N^2 - 1$$

Adjoint of $SU(N)$ has $\dim N^2 - 1$

* T^a are normalized w.r.t to the Killing form

$$(T^a, T^b) = \text{Tr}(T^a T^b) = \delta^{ab}$$

$$[T^a, T^b] = if^{abc} T^c$$

-D Gauge fix using the Gervias-Neveu gauge

$$\mathcal{L}_{gf} = -\frac{1}{2} \text{Tr}(H_{\mu\nu})^2 \quad | \quad H_{\mu\nu} \equiv \partial_\mu A_\nu - \frac{ig}{\sqrt{2}} A_\mu A_\nu$$

Note

$$F_{\mu\nu} = H_{\mu\nu} - H_{\nu\mu}$$

and so

$$\begin{aligned} \mathcal{L}_{YM} &= -\frac{1}{4} \text{Tr}((H_{\mu\nu} - H_{\nu\mu})(H^{\mu\nu} - H^{\nu\mu})) \\ &= -\frac{1}{4} \text{Tr}(H_{\mu\nu} H^{\mu\nu} - H_{\mu\nu} H^{\nu\mu} - H_{\nu\mu} H^{\mu\nu} - H_{\nu\mu} H^{\nu\mu}) \\ &= -\frac{1}{2} \text{Tr}(H_{\mu\nu} H^{\mu\nu} - H_{\mu\nu} H^{\nu\mu}) \end{aligned}$$

~> Now fix a gauge: $G(x)$: matrix valued and now add

$$\mathcal{L}_{gf} = -\frac{1}{2} \text{Tr}(G G)$$

to $\mathcal{L}_{YM}$: ignore gauge parameter ξ as well

as ghost is at TREE LEVEL

Choose $G = H_m^M = \partial^M A_m - \frac{ig}{\sqrt{2}} A^M A_m$

Note G is non-hermitian (from i) and hence $\mathcal{L}_{gf}^\dagger \neq \mathcal{L}_{gf}$ which is okay?

$\mathcal{L}_{tot} = \mathcal{L}_{YM} + \mathcal{L}_{gf}$

$\mathcal{L} = -\frac{1}{2} \text{Tr} (H_{MN} H^{MN} - H_{mN} H^{mN} + H_m^N H_N^m)$

Cancelling terms via int. by parts yields

$\mathcal{L} = \text{Tr} \left(-\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu - i\sqrt{2} \partial^M A^N A^M A_N + \frac{g^2}{4} A^M A^N A_M A_N \right)$

Feynmann Rules:

gluon - gluon prop $\langle \Omega | A_x^a A_y^b | \Omega \rangle = \int \frac{d^4 p}{(2\pi)^4} \frac{\delta^{ab} \eta_{\mu\nu}}{p^2 - i\epsilon} e^{-ip(x-y)}$

$\Delta_{\mu\nu}^{ab}(k^2) = \frac{\delta^{ab} \eta_{\mu\nu}}{k^2}$

Color factors: 4-gluon tree amplitudes

$C_s \equiv \tilde{f}^{a_1 a_2 b} \tilde{f}^{b a_3 a_4}$

$C_t \equiv \tilde{f}^{a_1 a_3 b} \tilde{f}^{b a_4 a_2}$

s-channel

t-channel

$C_u \equiv \tilde{f}^{a_1 a_4 b} \tilde{f}^{b a_2 a_3}$

u-channel

so we have $s + t + u = 4m^2$, in massless case $s + t + u = 0$

By Jacobi:

$C_s + C_t + C_u = 0$

there are only two independent color structures for tree level 4-gluon comp.

$$if^{abc} = \text{Tr}(T^a T^b T^c) - \text{Tr}(T^b T^a T^c)$$

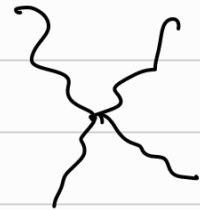
Proof:

$$\begin{aligned} \text{Tr}(T^a T^b T^c) &= if^{abd} \text{Tr}(T^d T^c) + \text{Tr}(T^b T^a T^c) \\ \text{Tr}(T^a T^b T^c) &= if^{abd} \text{Tr}(T^d T^c) + \text{Tr}(T^b T^a T^c) \\ if^{abc} &= \text{Tr}(T^a T^b T^c) - \text{Tr}(T^b T^a T^c) \end{aligned}$$

* Completeness Relation:

$$(T^a)_i^j (T^a)_k^l = \delta_i^l \delta_k^j - \frac{1}{N} \delta_i^j \delta_k^l$$

• S-channel 4-gluon diagram



$$i^2 \tilde{f}^{a_1 a_2 b} \tilde{f}^{b a_3 a_4}$$

$$= (\text{Tr}(T^{a_1} T^{a_2} T^b) - \text{Tr}(T^{a_2} T^{a_1} T^b)) (\text{Tr}(T^b T^{a_3} T^{a_4}) - \text{Tr}(T^{a_3} T^b T^{a_4}))$$

(no first term)

$$\text{Tr}(T^{a_1} T^{a_2} T^b) \text{Tr}(T^b T^{a_3} T^{a_4})$$

$$(T^{a_1})_k^l (T^{a_2})_j^i (T^b)_l^j (T^b)_n^m (T^{a_3})_p^n (T^{a_4})_m^p$$

$$(T^b)_l^j (T^b)_n^m = \delta_l^m \delta_n^j - \frac{1}{N} \delta_l^j \delta_n^m$$

$$\begin{aligned} & (T^{a_1})_k^l (T^{a_2})_j^i (\delta_l^m \delta_n^j) (T^{a_3})_p^n (T^{a_4})_m^p \\ & - (1/N) (T^{a_1})_k^l (T^{a_2})_j^i \delta_l^j \delta_n^m (T^{a_3})_p^n (T^{a_4})_m^p \end{aligned}$$

$$= (T^{a_1})_k^l (T^{a_2})_j^i (T^{a_3})_p^n (T^{a_4})_m^p$$

$$- (1/N) (T^{a_1})_k^l (T^{a_2})_j^i (T^{a_3})_p^n (T^{a_4})_m^p$$

$$= \text{Tr}[T^{a_1} T^{a_2} T^{a_3} T^{a_4}] - \frac{1}{N} \delta^{a_1 a_2} \delta^{a_3 a_4}$$

(second term)

$$= \text{Tr}(T^{a_2} T^{a_1} T^b) \text{Tr}(T^b T^{a_3} T^{a_4})$$

$$= -\text{Tr}(T^{a_2} T^{a_1} T^{a_3} T^{a_4}) + \frac{1}{N} \delta^{a_2 a_1} \delta^{a_4 a_3}$$

(Third term)

$$= \text{Tr}(T^{a_1} T^{a_2} T^b) \text{Tr}(T^b T^{a_4} T^{a_3})$$

$$= -\text{Tr}(T^{a_1} T^{a_2} T^{a_4} T^{a_3}) + \frac{1}{N} \delta^{a_1 a_2} \delta^{a_4 a_3}$$

(Fourth term)

$$\text{Tr}(T^{a_2} T^{a_1} T^b) \text{Tr}(T^b T^{a_4} T^{a_3})$$

$$= \text{Tr}(T^{a_2} T^{a_1} T^{a_4} T^{a_3}) = \frac{1}{N} \delta^{a_2 a_1} \delta^{a_4 a_3}$$

* All $\frac{1}{N}$ terms cancel and we are left with

$$f^{a_1 a_2 b} f^{b a_3 a_4} = -\text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4}) + \text{Tr}(T^{a_2} T^{a_1} T^{a_3} T^{a_4})$$

$$- \text{Tr}(T^{a_2} T^{a_1} T^{a_4} T^{a_3}) + \text{Tr}(T^{a_1} T^{a_2} T^{a_4} T^{a_3})$$

4-gluon tree amplitude

$$A_n^{\text{full, tree}} = g^2 (A_n[1234] \text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4}) + \text{-(perm}(234)\text{)})$$

$A_n[1234]$: color ordered (partial) amplitudes

* $A_n[\]$ are gauge invariant

→ Color structure generalizes to an n-pt tree level amp. involving any particles in the adjoint of G.

$$A_n^{\text{full, tree}} = g^{n-2} \sum_{\text{perms } \sigma} A_n[1 \sigma(2 \dots n)] \text{Tr}(T^{a_1} T^{\sigma(a_2)} \dots T^{a_n})$$

ex: 5pt f^{μ} will have g^2 from A^4 and a factor of g from $\partial^{\mu} A^{\nu} A_{\mu} A_{\nu}$.

Vertex Rules:

(3-gluon vertex): $\langle \Omega | A^{\mu_1} A^{\mu_2} A^{\mu_3} | \Omega \rangle$

$$= i(-i\sqrt{2}g) \int d^4\omega \langle \Omega | A^{\mu_1} A^{\mu_2} A^{\mu_3} \partial^\sigma A_\omega^\lambda A_\lambda^\omega A_\sigma^\omega | \Omega \rangle$$

In the Canonical Quant. case

$$A^\lambda(\omega) = \sum_{\lambda=1}^3 \int d^3k \left[\epsilon_\lambda^*(\vec{k}) a_\lambda(k) e^{ikx} + \epsilon_\lambda(k) a_\lambda^\dagger e^{-ikx} \right]$$

$$\begin{aligned} \partial_\omega^\sigma \langle \Omega | A^{\mu_2} A_\omega^\lambda | \Omega \rangle &= \partial_\omega^\sigma \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip(x-\omega)} g^{\mu_2\lambda}}{p^2 - i\epsilon} \\ &= \int \frac{d^4p}{(2\pi)^4} \frac{(-ip^\sigma) e^{-ip(x-\omega)} g^{\mu_2\lambda}}{p^2 - i\epsilon} \end{aligned}$$

$\Rightarrow$ obtain $(-ip^\mu)$ for $\partial^\mu A^\nu A_\nu A_\mu$

Thus, $\underbrace{A_1 A_2 A_3 \partial^\mu A^\nu A_\nu A_\mu}_{1 \leftrightarrow 2} + 1 \leftrightarrow 2$

$$\begin{aligned} & i(-i\sqrt{2}g)(-ip^\mu \Delta^{\nu\mu_3}) g_{\mu_2\nu} g_{\mu_1\mu_3} + 1 \leftrightarrow 2 \\ & -i\sqrt{2}g p_{\mu_1} \delta_{\mu_2}^{\mu_3} -i\sqrt{2}g p_{\mu_2} \delta_{\mu_1}^{\mu_3} \\ & -i\sqrt{2}g p_{\mu_1} g_{\mu_3\mu_2} -i\sqrt{2}g p_{\mu_2} g_{\mu_1\mu_3} \\ & -i\sqrt{2}g p_{\mu_3} g_{\mu_1\mu_2} -i\sqrt{2}g p_{\mu_2} g_{\mu_1\mu_3} \\ & -i\sqrt{2}g p_{\mu_1} g_{\mu_2\mu_3} -i\sqrt{2}g p_{\mu_3} g_{\mu_2\mu_1} \end{aligned}$$

$$= -i\sqrt{2}2 (p_\mu g_{\mu_2\mu_3} + p_{\mu_2} g_{\mu\mu_3} + p_{\mu_3} g_{\mu\mu_2})$$

I'm assuming we multiply this by $1/2$ for a symmetry factor....

(3-point Gluon Vertex): 

$$V^{\mu_1 \mu_2 \mu_3} (p_1, p_2, p_3) = -\sqrt{2} (p_1^{\mu_3} \eta^{\mu_1 \mu_2} + p_2^{\mu_1} \eta^{\mu_2 \mu_3} + p_3^{\mu_2} \eta^{\mu_1 \mu_3})$$

4 point Gluon Vertex

$$V^{\mu_1 \mu_2 \mu_3 \mu_4} (p_1, p_2, p_3, p_4) = \eta^{\mu_1 \mu_3} \eta^{\mu_2 \mu_4}$$

* For all outgoing momenta $E_{in \pm}(p_i)$ is needed for each external Gluon.

$$A_3[123] = -\sqrt{2} [(E_1 E_2)(E_3 P_1) + (E_1 E_3)(E_2 P_3) + (E_2 E_3)(E_1 P_2)]$$

$$A_4[1234] = (E_1 E_3)(E_2 E_4)$$

* In our notation: we have no crossing diagrams



Label each particle counter clock wise around the vertex.

$$\text{* Recall } \epsilon_+^\mu(p, q) = \frac{-\langle q | \gamma^\mu | p \rangle}{\sqrt{2} \langle q p \rangle} \quad \epsilon_-^\mu = \frac{-\langle q | \gamma^\mu | p \rangle}{\langle q p \rangle}$$

$$\epsilon_{+i}^\mu \epsilon_{+j}^\mu = \frac{1}{2} \frac{1}{\langle q_i p_i \rangle \langle q_j p_j \rangle} \langle q_i | \gamma^\mu | p_i \rangle \langle q_j | \gamma_\mu | p_j \rangle$$

$$= \frac{\langle q_i q_j \rangle [p_i p_j]}{\langle q_i p_i \rangle \langle q_j p_j \rangle}$$

$$\epsilon_{+i}^\mu \epsilon_{-j}^\mu = \frac{\langle q_i p_j \rangle [q_i p_i]}{\langle q_i p_i \rangle \langle q_j p_j \rangle}$$

∴

$$\epsilon_i^\pm \epsilon_j^\pm = \frac{\langle q_i q_j \rangle [p_i p_j]}{\langle q_i p_i \rangle \langle q_j p_j \rangle} \quad \epsilon_i^\pm \epsilon_j^\mp = \frac{\langle p_i p_j \rangle [q_i q_j]}{[q_i p_i] [q_j p_j]}$$

$$E_+ E_- = \frac{\langle q_i p_j \rangle [q_j p_i]}{\langle q_i p_i \rangle [q_j p_j]}$$

Which we now use. Take $A_3[123]$ with

$$A_3[1^- 2^- 3^+] = -\sqrt{2} (p_1 \epsilon_3^+ (\epsilon_1^- \epsilon_2^-) + (p_2 \epsilon_1^+) (\epsilon_2^- \epsilon_3^+) + (p_3 \epsilon_2^+) (\epsilon_1^- \epsilon_3^+))$$

$$(p_i)_\mu \epsilon_j^{+\mu} = -\frac{1}{\sqrt{2}} \frac{\langle q | p_i | p_j \rangle}{\langle q p_j \rangle} = \frac{\langle q p_i \rangle [p_i p_j]}{\sqrt{2} \langle q p_j \rangle}$$

$$(p_i)_\mu \epsilon^{+\mu} = \frac{[q p_i] \langle p_i p_j \rangle}{\sqrt{2} [q p_j]}$$

Plugging these in we find

$$p_1 \epsilon_3^+ (\epsilon_1^- \epsilon_2^-) = \left(\frac{\langle q_3 p_1 \rangle [p_1 p_3]}{\sqrt{2} \langle q_3 p_3 \rangle} \right) \left(\frac{\langle p_1 p_2 \rangle [q_1 q_2]}{[q_1 p] [q_2 p_2]} \right)$$

$$p_2 \epsilon_1^+ (\epsilon_2^- \epsilon_3^+) = \left(\frac{[q_1 2] \langle 12 \rangle}{\sqrt{2} [q_1 1]} \right) \left(\frac{\langle q_3 2 \rangle [q_2 3]}{\langle q_2 3 \rangle [q_2 2]} \right)$$

$$p_3 \epsilon_2^+ (\epsilon_1^- \epsilon_3^+) = \left(\frac{[q_2 3] \langle 23 \rangle}{\sqrt{2} [q_2 2]} \right) \left(\frac{\langle q_3 1 \rangle [q_1 3]}{[q_1 1] \langle q_2 3 \rangle} \right)$$

$$A_3[1^- 2^- 3^+] = \left\{ -\langle 12 \rangle [q_1 q_2] [13] \langle q_3 1 \rangle - \langle 12 \rangle [q_1 2] \langle q_3 2 \rangle [q_2 3] - \langle 23 \rangle [q_2 3] \langle q_3 1 \rangle [q_1 3] \right\} / \langle q_3 3 \rangle [q_2 2] [q_1 1]$$

• By 3-particle kinematics

$$(i) |1\rangle \propto |2\rangle \propto |3\rangle$$

then all $\langle 12 \rangle, \langle 23 \rangle$ vanish...

(ii) $|1\rangle \propto |2\rangle \propto |3\rangle$: only the first term vanishes

$$A_3[1^- 2^- 3^+] = - \left(\frac{\langle 12 \rangle [2q_1] \langle 2q_3 \rangle [q_2 3] + \langle 23 \rangle [q_2 3] \langle q_3 1 \rangle [q_1 3]}{\langle q_3 3 \rangle [q_2 2] [q_1 1]} \right)$$

• Can now use $\langle 12 \rangle [2q_1] = -\langle 13 \rangle [3q_1] = \langle 13 \rangle [q_1 3]$

$$= [2q_1] [q_2 3] \langle 12 \rangle \langle 2q_3 \rangle + \langle 23 \rangle [q_2 3] \langle q_3 1 \rangle [q_1 3]$$

$$\frac{\langle 341 \rangle \langle 423 \rangle}{\langle 433 \rangle [q_2 2] [q_1 1]} \left(\langle 13 \rangle \langle 243 \rangle + \langle 23 \rangle \langle 431 \rangle \right)$$

But shorten

$$\begin{aligned} \langle 431 \rangle \langle 23 \rangle + \langle 432 \rangle \langle 31 \rangle + \langle 433 \rangle \langle 12 \rangle &= 0 \\ (\langle 431 \rangle \langle 23 \rangle + \langle 13 \rangle \langle 243 \rangle) &= -\langle 12 \rangle \langle 433 \rangle \\ &= \frac{\langle 12 \rangle \langle 433 \rangle [3q_1] [q_2 3]}{\langle 433 \rangle [q_2 2] [q_1 1]} \end{aligned}$$

$$\begin{aligned} A_3 [1^- 2^- 3^+] &= \frac{\langle 433 \rangle [q_2 2] [q_1 1]}{\langle 12 \rangle [q_1 3] [q_2 3]} \\ &= \frac{\langle 12 \rangle^2}{[q_2 2] [q_1 1]} \end{aligned}$$

• Now use momentum cons.

$$\langle 21 \rangle [1q_1] = -\langle 23 \rangle [3q_1]$$

$$[q_1 3] / [q_1 1] = \langle 12 \rangle / \langle 23 \rangle$$

$$\text{and } \langle 13 \rangle [3q_2] = -\langle 12 \rangle [2q_2] = \langle 12 \rangle [q_2 2]$$

$$[q_2 3] / [q_2 2] = -\langle 12 \rangle / \langle 13 \rangle$$

$$\therefore A_3 [1^- 2^- 3^+] = \frac{\langle 12 \rangle^2}{\langle 23 \rangle \langle 31 \rangle}$$

Via Complex Conjugate:

$$A_3 [1^+ 2^+ 3^-] = \frac{[12]^2}{[23][31]}$$

*Note from $E_+ E_+$, $E_- E_-$, $E_+ E_-$ we can make all products vanish with the right choice of u :

Claim: If all or all but 1 of the external gluons have positive (+) helicity all possible polarization products are zero.

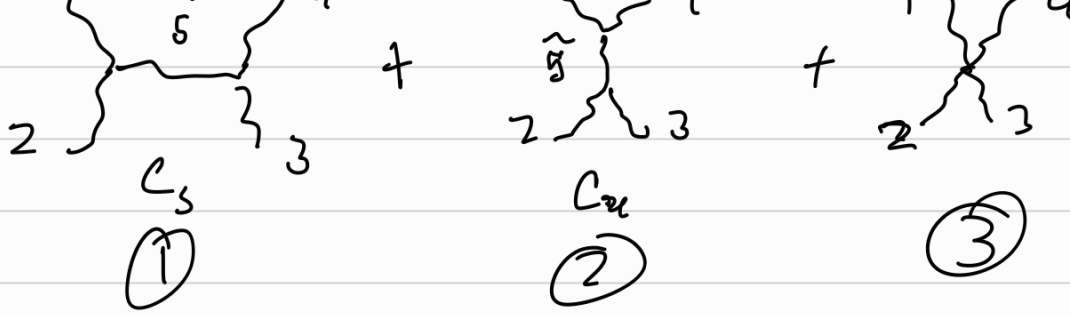
$$A(1^+, 2^+, \dots, n^+) = 0$$

$$A(1^+, 2^-, \dots, n^-) = 0$$

*Consider $A[1^-, 2^-, 3^+, 4^+]$

Must look at the contributing Feyn. diagrams

$$1) \quad 2) \quad 3) \quad 4) \quad 5) \quad 6)$$



$$\begin{aligned}
 (3) \quad (\epsilon_1^- \epsilon_3^+) (\epsilon_2^- \epsilon_4^+) &= (\epsilon_3^+ \epsilon_1^-) (\epsilon_4^+ \epsilon_2^-) \\
 \epsilon_4^+ \epsilon_2^- &= \left(\frac{1}{2} \right) \left(\frac{\langle q_4 | \gamma^\mu | q_2 \rangle}{\langle q_4 q_2 \rangle} \right) \left(\frac{[q_2 | \gamma_\mu | q_4]}{[q_2 q_4]} \right) \\
 &= \frac{\langle q_4 q_2 \rangle [q_4 q_2]}{\langle q_4 q_2 \rangle [q_2 q_4]}
 \end{aligned}$$

$$\therefore (1) = \left(\frac{\langle q_4 q_2 \rangle [q_4 q_2]}{\langle q_4 q_2 \rangle [q_2 q_4]} \right) \left(\frac{\langle q_3 q_1 \rangle [q_3 q_1]}{\langle q_3 q_1 \rangle [q_1 q_3]} \right)$$

→ Make vanish: $q_1 = q_2 = p_3$
 $q_3 = q_4 = p_2$

i.e. opposes so that the denominator does not vanish.

With this choice: $(3) = 0$

$$(2) \quad (iV_{\mu_2 \mu_3 \mu_5} (235)) \frac{i g^{\mu_2 \mu_3} \tilde{\mu}_5}{s_{14}} (iV_{\mu_1 \mu_4 \tilde{\mu}_5} (145)) \times (\epsilon^{\mu_1} \epsilon^{\mu_2} \epsilon^{\mu_3} \epsilon^{\mu_4})$$

$$\rightarrow g^{\mu_2 \mu_3} \epsilon_-^{\mu_2} V_{\mu_2 \mu_3 \mu_5} (235) \epsilon_+^{\mu_3}$$

$$g^{\mu_2 \mu_3} \epsilon_-^{\mu_2} (-\sqrt{2}) (\mathcal{H}_{\mu_2 \mu_3} (p_2)_{\mu_5} + (p_3)_{\mu_2} \mathcal{H}_{\mu_3 \mu_5} + (p_5)_{\mu_5} \mathcal{H}_{\mu_2 \mu_3})$$

$$(-\sqrt{2}) (\epsilon^2 \epsilon^3) (p_2)^{\mu_5} + ((p_3) \epsilon^2)$$

$$\dots i g^2 A(1-2-3+4) = (-i\sqrt{2} g^2) \left(\frac{i}{s_{12}} \right) (\epsilon_1 \epsilon_4 k_5 \epsilon_2 k_4 \epsilon_3)$$

$$\dots \frac{\langle 12 \rangle^4}{A(1-2-3+4)} = \dots$$

$$A_n[1^+ 2^- 3^+ 4^-] = \langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle$$

* Parke-Taylor n-gluon tree Amplitude:

$$A_n[1^+ \dots i^- \dots j^- \dots n^+] = \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$$

Diagrams which contribute to an n-gluon tree amplitude:

n =	3	4	5	6	7
# diagrams	1	3	10	38	154

Catalan numbers

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{2n!}{n! (n+1)!}$$

C_1 corresponds to $n=3$,
 C_2 to $n=4$, etc...

* We can take $\frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$

$$A_4[1^+ 2^- 3^+ 4^-] = \langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle$$

and it express it as $[\]$

-D

$$\langle 21 \rangle [14] = -\langle 23 \rangle [34]$$

$$\langle 12 \rangle / \langle 23 \rangle = [34] / [14]$$

$$\leadsto \frac{\langle 12 \rangle^2 [34]}{[14] \langle 34 \rangle \langle 41 \rangle} \quad \begin{array}{l} \langle 12 \rangle [23] = -\langle 14 \rangle [43] \\ \langle 12 \rangle / \langle 41 \rangle = [43] / [23] \end{array}$$

$$\leadsto \frac{\langle 12 \rangle [34]^2}{[14] \langle 34 \rangle [23]} \quad \begin{array}{l} (p_1 + p_2)^2 = (p_3 + p_4)^2 \\ p_1 \cdot p_2 = p_3 \cdot p_4 \\ \langle 12 \rangle [12] = \langle 34 \rangle [34] \end{array}$$

$$\therefore A_4[1^+ 2^- 3^+ 4^-] = \frac{[34]^3}{[12][23][41]}$$

as claimed

Color ordering of Amplitudes:

(i) Cyclic from the trace structure

$$A_n[12 \dots n] = A_n[2 \dots n1]$$

(ii) Reflection

$$A_n[22 \dots n] = (-1)^n A_n[n \dots 21]$$

(iii) $U(1)$ decoupling identity

$$A_n[123\dots n] + A_n[213\dots n] + A_n[231\dots n] + \dots + A_n[23\dots 1n] = 0$$

$\leadsto$ (iii) is also called the photon decoupling identity.

- (ii) Not too sure how to convince myself of this
- (iii) Note $\text{Tr}(T^a) = 0$, then, if T^a , $a = 1, \dots, N^2$ spans a basis for $GL(N)$, we can take

$$a = N^2 \text{ as } T^{N^2} = N^{-1/2} \mathbb{I}$$

$\leadsto$ Completeness relation: (Since $\{T^a\}$ form a complete set)

$$(T^a)_i^j (T^a)_k^l \propto \delta_i^l \delta_k^j$$

Note

$$\text{Tr}(T^a T^a) = \delta^{aa} = N^2 = (T^a)_i^j (T^a)_j^i$$

$\therefore$ Constant is 1:

$$(T^a)_j^i (T^a)_l^k = \delta_l^k \delta_j^i \quad a = 1, \dots, N^2$$

Now, from a traceless T^a : take $T^{N^2} = N^{-1/2} \mathbb{I}$

$$\Rightarrow (T^a)_j^i (T^a)_l^k + \delta_j^i \delta_l^k \frac{1}{N} = \delta_l^k \delta_j^i$$

$$(T^a)_j^i (T^a)_l^k = \delta_l^k \delta_j^i - \frac{1}{N} \delta_j^i \delta_l^k$$

- Taking $T^a \propto \mathbb{I}$, then $SU(N) \rightarrow U(N)$

$T^a \sim U(1)$ generator commutes w all others

T^a does not appear in

$$[A_m, A_n]$$

$\Rightarrow$ All scattering amp. will vanish if they contain T^a

$\leadsto$ Use $(T^a)_j^i (T^a)_l^k = \delta_l^k \delta_j^i$ since all terms with T^a will vanish

ex: $n=4$: $T^{a_4} \propto \mathbb{I}$ in

$$T = g^{n-2} \leq \text{Tr}(T^{a_1} \dots T^{a_{n-1}}) A[1 \dots n]$$

$$0 = \text{Tr}(1234) A[1234] + \text{Tr}(1324) A[1324] + \text{Tr}(1342) \\ + \text{Tr}(1432) + \text{Tr}(1423) + \text{Tr}(1243)$$

$$\therefore A[1234] \text{Tr}(123) + \text{Tr}(132) A[1324] + \\ \text{Tr}(132) A[1342] + \text{Tr}(132) A[1432] + \text{Tr}(123) A[1423] \\ + \text{Tr}(123) A[1243]$$

or

$$0 = \text{Tr}(123) \{ A[1234] + A[1243] + A[1423] \} \\ + \text{Tr}(132) \{ A[1324] + A[1342] + A[1432] \}$$

Both must be zero

$$A[1234] + A[1243] + A[1423] = 0 \quad (4)$$

$$A[1324] + A[1342] + A[1432] = 0 \quad (5)$$

$$(4) \quad A[1243] = A[3421] = A[1342]$$

$\therefore$

$$A[1423] + A[1243] + A[1234] = 0$$

and similarly for (5)

Can also show for explicit $A[1^2 2^{-3} n^1] = \frac{\langle 12 \rangle^4}{\langle 123 \rangle \langle 234 \rangle \langle 342 \rangle}$

$(n-1)!$: permutations of $(n-1)$ elements

ex: 3: abc, bac, bca, cba, cab, acb i.e. $6 = 3!$

* Thus, the trace Basis has $(n-1)!$ elements in $\sum_{\sigma} \text{Tr}(T^{a_1} T^{a_2} \dots T^{a_n}) A[1 \sigma(2 \dots n)]$

Kleiss-Kuijper relations:

$$A_n[1, \{\alpha\}, n, \{\beta\}] = (-1)^{|\beta|} \sum_{\sigma \in \text{OP}(\{\alpha\}, \{\beta\}^T)} A_n[1, \sigma, n]$$

$\{\beta^T\}$: reverse ordering of the labels $\{\beta\}$

Σ : over ordered permutations (OP)

and $\{\alpha\} \cup \{\beta^T\}$ is the ordering within $\{\alpha\}$

and $\{32\}$ is preserved

ex: 5-pt. amplitude

$$(HS) = A_5[1, \{2\}, 5, \{3, 4\}]$$

$$\Sigma_0 \propto U P^T = \{23\} U \{43\}$$

$$\sigma = \{243\}, \{423\}, \{432\}$$

$$\therefore A[12534] = A_5[12435] + A_5[14235] + A_5[14325]$$

* $n=4$

$$A_4[1, \{2\}, 4, \{3\}] = (-1)(A[1234] + A[1324])$$

or

$$A[1243] + A[1234] + A[1324] = 0$$

which is the $U(1)$ (photon) decoupling.

* $A_5[1, \{2, 3\}, 5, \{4\}]$

$$= (-1)A[12345] - A[12435] - A[14235]$$

$\therefore$

$$\rightarrow A[12354] + A[12345] + A[12435] + A[14235] = 0$$

Now, by $U(1)$ decoupling:

$$\rightarrow A[51234] + A[15234] + A[12534] + A[12354] = 0$$

$$A[12354] + A[12435] + A[14235]$$

$$- A[15234] - A[12534] - A[12354] = 0$$

$$\hookrightarrow A[52341] = (-1)A[14325]$$

$$A[12354] + A[12435] + A[14235] + A[14325]$$

$$- A[12534] - A[12354] = 0$$

and we obtain

$$A[12534] = A[12435] + A[14235] + A[14325]$$

(as eq. 2.87) (as claimed.)

* Kleiss-Kuijter relates cyclic, reflection &

$U(1)$ decoupling to reduce the # of independent

n -gluon tree amplitudes to $(n-2)!$

Recap: Color ordered amps:

1) Cyclic: $A_n[12\dots n] = A_n[2\dots n1]$

2) Reflection: $A_n[1\dots n] = (-1)^n A_n[n\dots 21]$

3) $U(1)$ decoupling:

$$A_n[12\dots n] + A_n[21\dots n] + \dots + A_n[23\dots 1n] = 0$$

• photon decoupling:

ex 2.27: $A[1^+ \dots i^- \dots j^+ \dots n^+] = \frac{\langle \epsilon_j \rangle^n}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$

$n=4$:

$$A_n[--++] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$$

(i) trivially true

(ii) "

(iii) $A[1234] + A[2134] + A[2314]$